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Inventor(s)	Martinez; Virginia et al.

Single-vector Type I vectors

Abstract

The invention relates to the production and use of Cas-encoding sequences and vectors comprising these. Aspects of the invention provide products, vectors, delivery vehicles, uses and methods for producing Cas-encoding sequences in bacterial or archaeal cells.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/201,736, filed Nov. 27, 2018, which claims priority to Great Britain Patent Application No. 1816700.7, filed Oct. 14, 2018, and Great Britain Patent Application No. 1817509.1, filed Oct. 27, 2018, the contents of each of which are hereby incorporated herein by reference in their entirety.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

(1) The contents of the electronic sequence listing (786212000603seqlist.xml; Size: 38,386 bytes; and Date of Creation: Oct. 27, 2023) is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The invention relates to the production and use of Cas-encoding sequences and vectors comprising these. Aspects of the invention provide products, vectors, delivery vehicles, uses and methods for producing Cas-encoding sequences in bacterial or archaeal cells.

BACKGROUND

(3) The state of the art describes vectors and uses of these that employ CRISPR/Cas systems. For example, reference is made to WO2017/118598, US20180140698, US20170246221, US20180273940, US20160115488, US20180179547, US20170175142, US20160024510, US20150064138, US20170022499, US20160345578, US20180155729, US20180200342, WO2017112620, WO2018081502, PCT/EP2018/066954, PCT/EP2018/066980, PCT/EP2018/071454 and U.S. Ser. No. 15/985,658 and equivalent publications by the US Patent and Trademark Office (USPTO) or WIPO, the disclosures of which are incorporated herein by reference.

SUMMARY OF THE INVENTION

(4) The invention provides the following configurations.

(5) In a First Configuration

(6) A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

(7) In an example, the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

(8) The invention also provides a delivery vehicle comprising the vector, as well as a pharmaceutical composition comprising the vector or vehicle and a pharmaceutically acceptable diluent, excipient or carrier.

(9) The invention also provides a method of treating or reducing the risk of a disease or condition in a human or animal subject, the method comprising administering the vector, vehicle or composition to the subject, and introducing the vector into target host bacterial or archaeal cells in the subject (eg, in a gut microbiota, lung, eye or blood of the subject), wherein the Cas cuts (or otherwise modifies) one or more target sequences in the target cells and the cells are killed or growth or proliferation of the cells is reduced.

(10) In a Second Configuration

(11) A method of amplifying copies of a DNA encoding a functional Cas protein (optionally a Cas nuclease) in a bacterial or archaeal production strain of cells, the method comprising (a) Providing production strain cells, each cell comprising a copy of said DNA, wherein each DNA comprises a nucleotide sequence encoding said Cas, wherein the nucleotide sequence is under the control of a promoter for controlling the expression of the Cas in the production strain cell, the DNA comprising an origin of replication that is operable in the cell for replication of the DNA; (b) Culturing the cells to allow replication of the DNA, whereby the DNA is amplified; and (c) Optionally isolating copies of the DNA, Optionally wherein the promoter is an attenuated constitutive promoter.

In a Third Configuration

(12) Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for enhancing the yield of amplified DNA produced by the production host cells.

(13) In a Fourth Configuration

(14) Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing toxicity of the Cas in the production strain.

(15) In a Fifth Configuration

(16) Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain.

(17) In a Sixth Configuration

(18) Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding

- a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for promoting production cell viability during the amplification of the DNA.
- (19) In a Seventh Configuration
- (20) Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing the occurrence of Cas cutting of DNA.
- (21) In an Eighth Configuration
- (22) A method for enhancing the yield of amplified copies of a DNA construct in a population of bacterial or archaeal production strain cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.
- (23) In a Ninth Configuration
- (24) A method for reducing toxicity of a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.
- (25) In a Tenth Configuration
- (26) A method for reducing mutation of a DNA construct encoding a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.
- (27) In an Eleventh Configuration
- (28) A method for promoting production cell viability of a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct comprised by the cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.
- (29) In a Twelfth Configuration
- (30) A method for reducing the occurrence of Cas nuclease cutting of a DNA construct in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-1C. Type I CRISPR-Cas system of *C. difficile* targeting *E. coli* MG1655. (FIG. 1A)

Layout of the CRISPR Guided Vector™, CGV™. Plasmid 1: pSC101 on, pBAD promoter (induced by arabinose), cas3 and cascade genes. Plasmid 2: pCloDF13 on, pTac promoter (induced by IPTG), CRISPR array. (FIG. 1B) Dilution series (10.sup.1-10.sup.6) of drop spots (5 µl) of *E. coli* MG1655 harboring the CGV on LB agar plates with and without inducers. (FIG. 1C) CRISPR induction killed 99.9% of the population (grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

(2) FIGS. 2A-2C. Type I CRISPR-Cas system of *C. difficile* targeting *E. coli* MG1655. (FIG. 2A) Layout of the CRISPR Guided Vector™, CGV™. pSC101 on, pTac promoter (induced by IPTG), CRISPR array, pBAD promoter (induced by arabinose), cas3 and cascade genes. (FIG. 2B) Dilution series (10.sup.1-10.sup.6) of drop spots (5 µl) of *E. coli* MG1655 harboring the CGV on SM agar plates with and without inducers. (FIG. 2C) CRISPR induction killed 99% of the population (grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

(3) FIGS. 3A-3B. Time-kill curves for *E. coli* MG1655 harboring the CGV. (FIG. 3A) CRISPR induction killed 99% of the population in 60 minutes (dashed line). Growth in absence of induction is shown in black lines. CRISPR/Cas was induced at time-point 0 and monitored until 120 minutes. (FIG. 3B) Dilution series (10.sup.1-10.sup.6) of drop spots (5 µl) on SM agar plates of *E. coli* MG1655 after 60 minutes of induction.

(4) FIGS. 4A-4B. Specific killing of *E. coli* MG1655 with type I-B CRISPR-Cas system of *C. difficile* in a synthetic microbial consortium. (FIG. 4A) Bacteria count of a synthetic population composed of three different strains. CRISPR was induced at time-point 0 and monitored for 60 minutes. Growth in absence of induction is shown in black. CRISPR induction prompted 1-log reduction in viable cells of target strain *E. coli* MG1655, while leaving *E. coli* Top10 and *L. lactis* NZ9000 populations intact (dark grey bars). (FIG. 4B) Dilution series (10.sup.1-10.sup.6) of drop spots (5 µl) of the bacterial community mixture after 60 minutes of induction. *E. coli* MG1655 grows selectively on BHI+streptomycin, *E. coli* Top10 on ampicillin, and *L. lactis* NZ9000 on chloramphenicol.

(5) FIGS. 5A-5B. Killing of *E. coli* MG1655 with type I-B CRISPR-Cas system of *C. difficile* in a synthetic microbial consortium compared to a pure culture of *E. coli* MG1655. (FIG. 5A) CRISPR induction generated 4-log reductions in viable cells of target strain *E. coli* MG1655, both in the pure culture and in the community mixture (grey bars). Growth in absence of induction is shown in black. (FIG. 5B) Dilution series of a pure culture of *E. coli* MG1655 and the bacterial community mixture on streptomycin plates with and without inducers.

(6) FIGS. 6A-6B. Type I CRISPR-Cas system of *E. coli* targeting *E. coli* MG1655. (FIG. 6A) Dilution series (10.sup.1-10.sup.6) of drop spots (5 µl) of *E. coli* MG1655 harboring the CGV on SM agar plates with and without inducers. (FIG. 6B) CRISPR induction killed 99% of the population (grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™ which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

DETAILED DESCRIPTION

(7) The invention relates to the production and use of Cas-encoding sequences and vectors comprising these. Aspects of the invention provide products, vectors, delivery vehicles, uses and methods for producing Cas-encoding sequences in bacterial or archaeal cells.

(8) An aspect of the invention provides for the control of expression of Cas and optionally also Cascade proteins from single vectors, such as by regulated use of Cas modules in an operon and/or using attenuated promoters.

(9) Concepts:

(10) An aspect of the invention provides nucleic acid vectors that are useful for introducing into

target host cells of any eukaryotic or prokaryotic species (eg, ex vivo or in vitro) for expressing Type I Cas and optionally other components of a Type I CRISPR/Cas system. Usefully, the vector may in some examples therefore provide a single-vector means for introducing a complete exogenous Type I CRISPR/Cas system into a target cell for modification (eg, cutting by Cas3) of DNA in the target cell. In an example, a chromosomal target sequence (ie, protospacer that is cognate with the Cas3) is modified. In another example, an episomal DNA sequence is modified, for example a plasmid sequence or a DNA that has been introduced into the cell. The latter may be useful in a recombineering method of the invention wherein exogenous DNA in the target cell is cut by the Cas3 and optionally this produces one or more recombinogenic ends for recombination of the cut DNA with a further DNA of interest, thereby producing a recombination product in the cell. For example, in such a recombineering method, the target cell is a recombinogenic *E. coli* cell, eg, comprising a red/ET system. In another example, the target cell is an undesired cell (eg, a cell of a species or strain that is pathogenic to humans or animals, such as a bacterial disease-causing species or strain) and the cutting by Cas3 kills the cell. This may be useful for treating or preventing an infection in a human or animal harbouring target cells. The provision of single-vector means that express minimally a Cas endonuclease (eg, Cas3), cognate accessory proteins (eg, Cascade proteins) and at least one CRISPR array (or nucleotide sequence encoding a guide RNA (eg, a single guide RNA)), wherein the Cas, accessory proteins and array (or nucleotide sequence) comprise a functional CRISPR/Cas system is more convenient and the inventors believe more efficient for introducing into a target cell and effecting CRISPR/Cas modification of a target sequence therein than the use of 2 or 3 or more separate vectors (eg, a vector encoding the Cas nuclease and a different vector encoding the accessory proteins, and possibly a further vector comprising the array (or gRNA-encoding nucleotide sequence) which all need to transform the target cell for the system to function). This may provide one or more benefits, therefore, such as simplifying delivery (and thus the design of delivery vehicles), simplifying construction of the vector and vehicle and/or providing for better cutting or killing efficiencies. Conveniently, an example of the invention therefore uses an operon for the coordinated expression in the target cells of the Cas and accessory proteins (and optional also the array or gRNA-encoding sequence(s)). Whilst not wishing to be bound by any particular theory, the introduction of a single vector (eg, using an operon) as per the invention may advantageously coordinate the expression of the Cas and accessory proteins (and optionally production of cRNAs or gRNAs) so that these are available to operate together without undue delay in the target cell. This may be important to tip the balance between, on the one hand the target cell using its endogenous anti-restriction, endogenous Cas or other endogenous mechanisms that seek out and degrade invading phage and DNA, and on the other hand efficient cell killing or deactivation of such mechanisms by the invading CRISPR components of the vector of the invention. In such an arms race, concerted and early operation of the CRISPR components in the cell are likely to be important to gain the upper hand and effect cell killing. The invention provides means to assist this.

(11) By way of example, the invention thus provides the following Concepts:—

(12) 1. A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

(13) 2. The vector of concept 1, wherein the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

(14) 3. The vector of concept 2, wherein (a) the first sequence is between the promoter and the second sequence in the operon; (b) the operon comprises no Cas-encoding nucleotide sequences

between the promoter and the first nucleotide sequence; and/or (c) the operon comprises (in 5' to 3' direction) the promoter, the first sequence and the second sequence.

(15) 4. The vector of any preceding concept, wherein each promoter is a constitutive promoter.

(16) 5. The vector of any one of concepts 1 to 3, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or lac repressor).

(17) 6. The vector of any one of concepts 1 to 3, wherein the promoter is inducible.

(18) 7. The vector of any preceding concept, wherein the first sequence is under the control of a medium strength promoter.

(19) 8. The vector of any preceding concept, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $0.5 > AS > 0.1$.

(20) 9. The vector of any preceding concept, wherein the first sequence (and optionally the second sequence) is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(21) 10. The vector of concept 9, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

(22) 11. The vector of any preceding concept, wherein the vector comprises an origin of replication that is operable in the host cell.

(23) 12. The vector of any preceding concept, wherein the vector comprises an origin of replication that is operable in a bacterial cell of a vector production strain, wherein the Cas3 is not operable in the production strain cell to target and cut a chromosomal sequence thereof.

(24) 13. The vector of concept 12, wherein the first sequence is under the control of a promoter that is capable of controlling expression of the Cas3 at a level that is not toxic to the production strain cell.

(25) 14. The vector of any preceding concept, wherein the vector is a high copy number vector.

(26) 15. The vector of any preceding concept, wherein the first nucleotide sequence or operon is comprised by a mobile genetic element.

(27) 16. The vector of any preceding concept, wherein the vector is devoid of a Cas adaption module.

(28) 17. The vector of any preceding concept, wherein the vector is devoid of nucleotide sequence encoding one, more or all of a Cas1, Cas2, Cas4, Cas6, Cas7 and Cas8.

(29) 18. The vector of any preceding concept, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas11, Cas7 and Cas8a1.

(30) 19. The vector of concept 18, wherein the vector comprises nucleotide sequence encoding Cas3' and/or Cas3''.

(31) 20. The vector of concept 19, wherein the nucleotide sequences encoding the Cas3' and/or Cas3'' are between the promoter and the sequence(s) recited in concept 18.

(32) 21. The vector of any one of concepts 18 to 20, wherein the host cell comprises a Type IA CRISPR array that is cognate with the Cas3.

(33) 22. The vector of any one of concepts 18 to 20, wherein the host cell comprises an endogenous Type IB, C, U, D, E or F CRISPR/Cas system.

(34) 23. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8b1, Cas7 and Cas5.

(35) 24. The vector of concept 23, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 23.

- (36) 25. The vector of concept 23 or 24, wherein the host cell comprises a Type IB CRISPR array that is cognate with the Cas3.
- (37) 26. The vector of concept 23 or 24, wherein the host cell comprises an endogenous Type IA, C, U, D, E or F CRISPR/Cas system.
- (38) 27. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas5, Cas8c and Cas7.
- (39) 28. The vector of concept 27, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 27.
- (40) 29. The vector of concept 27 or 28, wherein the host cell comprises a Type IC CRISPR array that is cognate with the Cas3.
- (41) 30. The vector of concept 27 or 28, wherein the host cell comprises an endogenous Type IA, B, U, D, E or F CRISPR/Cas system.
- (42) 31. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8U2, Cas7, Cas5 and Cas6.
- (43) 32. The vector of concept 31, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 31.
- (44) 33. The vector of concept 31 or 32, wherein the host cell comprises a Type IU CRISPR array that is cognate with the Cas3.
- (45) 34. The vector of concept 31 or 32, wherein the host cell comprises an endogenous Type IA, B, C, D, E or F CRISPR/Cas system.
- (46) 35. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas10d, Cas7 and Cas5.
- (47) 36. The vector of concept 35, wherein the vector comprises a nucleotide sequence encoding Cas3' and/or Cas3".
- (48) 37. The vector of concept 36, wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in concept 35.
- (49) 38. The vector of any one of concepts 35 to 37, wherein the host cell comprises a Type ID CRISPR array that is cognate with the Cas3.
- (50) 39. The vector of any one of concepts 35 to 37, wherein the host cell comprises an endogenous Type IA, B, C, U, E or F CRISPR/Cas system.
- (51) 40. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8e, Cas11, Cas7, Cas5 and Cas6.
- (52) 41. The vector of concept 40, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 40.
- (53) 42. The vector of concept 40 or 41, wherein the host cell comprises a Type IE CRISPR array that is cognate with the Cas3.
- (54) 43. The vector of concept 40 or 41, wherein the host cell comprises an endogenous Type IA, B, C, D, U or F CRISPR/Cas system.
- (55) 44. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8f, Cas5, Cas7 and Cas6f.
- (56) 45. The vector of concept 44, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 44, wherein the vector is devoid of nucleotide sequence encoding further Cas between the promoter and the sequence encoding the Cas3.
- (57) 46. The vector of concept 44 or 45, wherein the host cell comprises a Type IF CRISPR array that is cognate with the Cas3.
- (58) 47. The vector of concept 44 or 45, wherein the host cell comprises an endogenous Type IA, B, C, D, U or E CRISPR/Cas system.
- (59) 48. The vector of any one of concepts 1 to 17, wherein the Cas and Cascade are (a) Type IA

Cas and Cascade proteins; (b) Type IB Cas and Cascade proteins; (c) Type IC Cas and Cascade proteins; (d) Type ID Cas and Cascade proteins; (e) Type IE Cas and Cascade proteins; (f) Type IF Cas and Cascade proteins; or (g) Type IU Cas and Cascade proteins.

(60) 49. The vector of any preceding concept, wherein the Cas and Cascade are *E. coli* (optionally Type IE or IF) Cas and Cascade proteins.

(61) 50. The vector of concept 49, wherein the *E. coli* is ESBL-producing *E. coli* or *E. coli* ST131-O25b:H4.

(62) 51. The vector of any preceding concept, wherein the Cas and Cascade are (a) *Clostridium* (eg, *C. difficile*) Cas and Cascade proteins, optionally *C. difficile* resistant to one or more antibiotics selected from aminoglycosides, lincomycin, tetracyclines, erythromycin, clindamycin, penicillins, cephalosporins and fluoroquinolones; (b) *Pseudomonas aeruginosa* Cas and Cascade proteins, optionally *P. aeruginosa* resistant to one or more antibiotics selected from carbapenems, aminoglycosides, cefepime, ceftazidime, fluoroquinolones, piperacillin and tazobactam; or (c) *Klebsiella pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*) Cas and Cascade proteins.

(63) 52. The vector of any preceding concept, wherein the Cas and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus* or *B. halodurans* Cas and Cascade proteins.

(64) 53. The vector of any preceding concept, wherein the Cas3 is a Cas3 of a CRISPR/Cas locus of a first bacterial or archaeal species, wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter is further than the distance between the Cas3-encoding sequence and the respective promoter comprised by the vector.

(65) 54. The vector of any preceding concept, wherein the distance between the promoter and the Cas3-encoding sequence and/or Cascade protein-encoding sequence(s) is shorter than in a corresponding wild-type Type I locus.

(66) 55. The vector of any preceding concept, wherein the vector comprises (i) a CRISPR array for producing crRNAs in the host cell and/or (ii) one or more nucleotide sequences encoding one or more guide RNAs (gRNAs or single gRNAs), wherein the crRNAs or gRNAs are cognate to the Cas3 (and optionally cognate to the Cascade proteins).

(67) 56. The vector of concept 55 when dependent from concept 2, wherein the array or gRNA-encoding sequence(s) are comprised by the operon and under the control of the promoter.

(68) 57. The vector of concept 56, wherein the array or gRNA-encoding sequence(s) are under the control of a promoter that is different from the promoter that controls the expression of the Cas3.

(69) 58. The vector of concept 56 or 57, wherein one or more of the crRNAs or gRNAs comprises a spacer sequence that is capable of hybridising to a target nucleotide sequence of the host cell, wherein the target sequence is adjacent a PAM, the PAM being cognate to the Cas3.

(70) 59. The vector of concept 58, wherein the target sequence is a chromosomal sequence of the host cell.

(71) 60. The vector of concept 58 or 59, wherein the Cas3 is operable to cut the target sequence.

(72) 61. The vector of any preceding concept, wherein the vector is a plasmid or phagemid.

(73) 62. A delivery vehicle comprising the vector of any preceding concept, wherein the delivery vehicle is capable of delivering the vector into the host cell.

(74) 63. The vehicle of concept 62, wherein the delivery vehicle is a phage, non-replicative transduction particle, nanoparticle carrier, bacterium or liposome.

(75) 64. The vector or vehicle of any preceding concept, wherein the host cell is a bacterial or archaeal cell, optionally, the host cell is a *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cell.

(76) 65. The vector or vehicle of any preceding concept for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

(77) 66. The vector or vehicle of concept 65, wherein the disease or condition is an infection of the subject with host cells (eg, bacterial cells), or wherein the disease or condition is mediated by host cells (eg, bacterial cells).

(78) 67. A pharmaceutical composition comprising the vector or vehicle of any preceding concept and a pharmaceutically acceptable diluent, excipient or carrier.

(79) 68. A method of treating or reducing the risk of a disease or condition in a human or animal subject, the method comprising administering the vector, vehicle or composition of any preceding concept to the subject, and introducing the vector into target host bacterial or archaeal cells in the subject (eg, in a gut microbiota, lung, eye or blood of the subject), wherein the Cas cuts (or otherwise modifies) one or more target sequences in the target cells and the cells are killed or growth or proliferation of the cells is reduced.

(80) 69. The method of concept 68, wherein the target cells are cells of a disease pathogen species.

(81) 70. The method of concept 68 or 69, wherein the target cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S pneumoniae* or *S. aureus* cells.

Embodiments

(82) An aspect of the invention provides improved ways of amplifying DNA constructs in bacterial and archaeal production strain cells. For example, the DNA may be a high copy number plasmid or phagemid comprising a constitutive promoter for controlling the expression of one or more Cas proteins when the DNA has been introduced into a target host bacterial or host cell. It is desirable, according to an aspect of the invention, to consider attenuating the promoter activity during amplification of the DNA in the production strain. This is useful, since the inventors have found that Cas expression in production strains may be toxic to production strain cells, thereby reducing the yield of amplified DNA. Toxicity may be due, for example, to off-target cutting of the production strain chromosomal DNA when the Cas is a nuclease (such as Cas9 or Cas3) and/or due to relatively high levels of expression of the Cas in the cells. Additionally or alternatively, undesirably the Cas expression or activity may impose a selective pressure that favours mutation and propagation of mutated DNA constructs (such as mutation in one more or all of a CRISPR/Cas operon, Cas-encoding gene, Cascade-encoding gene, CRISPR array and gRNA-encoding sequence of the DNA construct) in production cells, thereby reducing the yield of desired amplified constructs and imposing an undesired step of separating desired from mutated DNA constructs for further formulation into useful compositions. Such compositions may be pharmaceutical compositions, herbicides, pesticides, environmental remediation compositions etc. In one example, the promoter attenuation in production strains is achieved by using a medium strength (not high or low) promoter to control the Cas-encoding nucleotide sequence of the DNA constructs. A medium level of Cas expression may be tolerable in the production strains, and yet once the DNA is subsequently introduced into target host cells the Cas is expressed at sufficiently high levels to produce desired activity to modify (eg, cut) target sequences in target cells. In an alternative, the invention uses a repressible promoter, wherein the promoter is repressed in production strain, but not repressed in target host cells. For example, aspects of the invention use a tetracycline repressor (tetR) expressed in production strain cells that represses the promoter.

(83) Thus, the yield can be enhanced by one or more of (a) reducing toxicity of the Cas in the production strain; (b) reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain; (c) promoting production cell viability during the amplification of the DNA; and (d) reducing the occurrence of Cas cutting of DNA (optionally cutting of production host cell chromosomal DNA or said DNA construct).

(84) To this end, the invention provides Embodiments as follows:—

(85) 1. A method of amplifying copies of a DNA encoding a functional Cas protein (optionally a Cas nuclease) in a bacterial or archaeal production strain of cells, the method comprising (a)

Providing production strain cells, each cell comprising a copy of said DNA, wherein each DNA comprises a nucleotide sequence encoding said Cas, wherein the nucleotide sequence is under the control of a promoter for controlling the expression of the Cas in the production strain cell, the DNA comprising an origin of replication that is operable in the cell for replication of the DNA; (b) Culturing the cells to allow replication of the DNA, whereby the DNA is amplified; and (c) Optionally isolating copies of the DNA, wherein the promoter is an attenuated constitutive promoter.

(86) In an example, promoter is a medium strength promoter. In another example, the promoter is repressed in the production strain cell. Hence, the promoter is an attenuated promoter in these examples.

(87) 2. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for enhancing the yield of amplified DNA produced by the production host cells.

(88) 3. The use of paragraph 2, wherein the use is for enhancing said yield by (a) reducing toxicity of the Cas in the production strain; (b) reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain; (c) promoting production cell viability during the amplification of the DNA; and/or (d) reducing the occurrence of Cas cutting of DNA (optionally cutting of production host cell chromosomal DNA or said DNA construct).

(89) 4. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing toxicity of the Cas in the production strain.

(90) 5. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain.

(91) 6. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for promoting production cell viability during the amplification of the DNA.

(92) 7. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing the occurrence of Cas cutting of DNA.

(93) 8. A method for enhancing the yield of amplified copies of a DNA construct in a population of bacterial or archaeal production strain cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

(94) 9. A method for reducing toxicity of a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of a

DNA construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

(95) 10. A method for reducing mutation of a DNA construct encoding a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

(96) 11. A method for promoting production cell viability of a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct comprised by the cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

(97) 12. A method for reducing the occurrence of Cas nuclease cutting of a DNA construct in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

(98) 13. The use of paragraph 5 or 7, or the method of paragraph 10 or 12, wherein the mutation or cutting is mutation or cutting of host cell chromosomal DNA or the construct DNA.

(99) 14. The method or use of any one of paragraphs 2 to 13, wherein the promoter is a constitutive promoter.

(100) 15. The method or use of any preceding paragraph, wherein the promoter is repressed in the production strain cells (optionally repressed by a tetracycline repressor or a lac repressor).

(101) 16. The method or use of paragraph 15, wherein the promoter is P.sub.LtetO-1, P.sub.LlacO-1 or a repressible homologue thereof.

(102) Other examples of suitable repressible promoters are Ptac (repressed by lacI) and the Leftward promoter (pL) of phage lambda (which is repressed by the λ CI repressor). In an example, the promoter comprises a repressible operator (eg, tetO or lacO) fused to a promoter sequence. The corresponding repressor is encoded by a nucleic acid in the production strain (eg, a chromosomally-integrated sequence or a sequence comprised by an episome) and the repressor is expressed during the DNA or vector amplification method of the invention, whereby the promoter controlling Cas expression is repressed. In delivery vehicles that are subsequently produced from isolated amplified DNA/vector, the vehicle is devoid of an expressible nucleotide sequence encoding the repressor, whereby the promoter is functional when the DNA/vector is introduced into a target host cell. For example, in the absence of the repressor the promoter is constitutively ON for expression of the Cas. The system is therefore primed to work once the DNA/vector is introduced into the host cells, and this effect can be enhanced further by using a high copy number DNA/vector comprising an origin of replication that is operable in the host cell. A high copy number vector or DNA is also desirable in the production strain cells for enhancing yield of the DNA/vector, and by use of an attenuated promoter as described herein (eg, medium strength promoter and/or repressed promoter in the production strain cells) one can minimize Cas toxicity whilst culturing to maximize amplification and thus yield of the DNA/vector.

(103) 17. The method or use of any preceding paragraph, wherein the promoter is a medium strength promoter.

(104) 18. The method or use of any preceding paragraph, wherein the promoter has an Anderson

Score (AS) of $0.5 > AS > 0.1$.

(105) 19. The method or use of any preceding paragraph, wherein the nucleotide sequence encoding said Cas is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(106) 20. The method or use of paragraph 19, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

(107) 21. The method or use of any preceding paragraph, wherein the nuclease is Cas3 and optionally the DNA or cell encodes cognate Cascade proteins.

(108) 22. The method or use of any one of paragraphs 1 to 20, wherein the Cas is a Cas9.

(109) 23. The method or use of any preceding paragraph, wherein the production strain cells comprise a helper phage genome that is inducible to produce phage coat proteins in the cells, wherein the method further comprises inducing production of the phage proteins and causing packaging of the amplified DNA into phage particles or non-self-replicative transduction particles, and further isolating the phage or transduction particles and optionally formulating the particles into a pharmaceutical composition for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

(110) 24. The method or use of paragraph 23, wherein the particles are capable of infecting target host cells in the subject and transducing the cells with the DNA, wherein the Cas and crRNAs (or guide RNAs, gRNAs) encoded by the DNA are expressed in the cells, the crRNAs or (gRNAs) being operable to guide the Cas to a target nucleotide sequence (optionally a chromosomal sequence) comprised by the cells, wherein the Cas cuts the target sequences in the cells, thereby killing host cells and treating or reducing the risk of the disease or condition.

(111) 25. The method or use of paragraph 24, wherein the host cells are bacterial or archaeal cells, optionally, the host cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

(112) 26. The method or use of any preceding paragraph, wherein each DNA is comprised by a high copy number plasmid or phagemid.

(113) 27. The method or use of any preceding paragraph, wherein the DNA construct comprises one or more nucleotide sequences for producing crRNAs or gRNAs that are operable for Cas nuclease targeting in target host cells.

(114) Paragraphs & Generally Applicable Features

(115) The invention provides the following Paragraphs, which are supported by the Examples below. Any features of the Concepts are combinable with any features of the Embodiments. Any features of the Concepts are combinable with any features of the Embodiments. Any features of the Paragraphs are combinable with any features of the Embodiments.

(116) Any cell herein (eg, a production strain cell or target host cell) may be a bacterial cell, archaeal cell, algal cell, fungal cell, protozoan cell, invertebrate cell, vertebrate cell, fish cell, bird cell, mammal cell, companion animal cell, dog cell, cat cell, horse cell, mouse cell, rat cell, rabbit cell, eukaryotic cell, prokaryotic cell, human cell, animal cell, rodent cell, insect cell or plant cell. Preferably, the cell is a bacterial cell. Alternatively, the cell is a human cell. Optionally, the production strain cell(s) and target host cell(s) are of the same phylum, order, family, genus, species or strain.

- (117) 1. A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3, wherein the sequence is under the control of a promoter comprised by the vector for expression of the Cas3 in the cell.
- (118) In an example, the vector is a DNA vector, eg, ssDNA vector or dsDNA vector.
- (119) 2. The vector of paragraph 1, wherein the vector comprises a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.
- (120) 3. The vector of paragraph 2, wherein the Cascade protein(s) are cognate with the Cas3.
- (121) In an example, the Cas3 is cognate with Cascade proteins encoded by the host cell and/or encoded by a second operon. Optionally, the second operon is comprised by the vector. Optionally, the second operon is comprised by a second vector that is capable of introducing the second operon into the host cell, whereby the Cas3 and Cascade proteins are expressed from the operons in the host cell and are operable with crRNA or gRNA to target the Cas to a host cell target sequence, wherein the Cas3 is capable of modifying the target sequence.
- (122) 4. The vector of paragraph 2 or 3, wherein the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.
- (123) The term “operon” is known to the skilled person such as relating to a functioning unit of DNA containing at least expressible 2 nucleotide sequences respectively encoding for an expression product (eg, a respective translatable mRNA), wherein the sequences are under common promoter control.
- (124) 5. The vector of paragraph 4, wherein the first sequence is between the promoter and the second sequence in the operon.
- (125) 6. The vector of paragraph 4 or 5, wherein the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence.
- (126) Optionally, the Cas3 is a Cas3 encoded by a CRISPR/Cas locus of a first bacterial or archaeal species, wherein in the locus the Cas3-encoding sequence is 3' of Cascade protein-encoding sequences (ie, the latter are between the Cas3 and the 5'-most promoter of the locus).
- (127) Optionally, the Cas3 is a ygcB protein (eg, wherein the production strain cell and/or host target cell is an *E. coli*).
- (128) Optionally, the Cascade proteins comprise or consist of cas5 (casD, csy2) cas6 (cas6f, cse3, casE) cas7 (csc2, csy3, cse4, casC) cas8 (casA, cas8a1, cas8b1, cas8c, cas10d, cas8e, cse1, cas8f, csy1).
- (129) Optionally herein the promoter and the Cas3-encoding sequence are spaced no more than 150, 100, 50, 40, 30, 20 or 10 bp apart, eg, from 30-45, or 30-40, or 39 or around 39 bp apart.
- (130) Optionally herein a ribosome binding site and the Cas3-encoding sequence are spaced no more than 20, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 4 or 3 bp apart, eg, from 10-5, 6 or around 6 bp apart.
- (131) 7. The vector of any one of paragraphs 4 to 6, wherein the operon comprises (in 5' to 3' direction) the promoter, the first sequence and the second sequence.
- (132) 8. The vector of any preceding paragraph, wherein each promoter is a constitutive promoter.
- (133) 9. The vector of any one of paragraphs 1 to 7, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or lac repressor).
- (134) 10. The vector of any one of paragraphs 1 to 7, wherein the promoter is inducible.
- (135) 11. The vector of any preceding paragraph, wherein the first sequence is under the control of a weak promoter.
- (136) 12. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a medium strength promoter.

(137) 13. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a strong promoter.

(138) In an example, the promoter is in combination with a Shine-Dalgarno sequence comprising the sequence 5'-aaagaggagaaa-3' (SEQ ID NO: 5) or a ribosome binding site homologue thereof.

(139) 14. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $AS \geq 0.5$.

(140) See Table 2 for more information on Anderson Scores in relation to promoters.

(141) 15. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $0.5 > AS > 0.1$.

(142) 16. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of ≤ 0.1 .

(143) 17. The vector of any one of paragraphs 1 to 7, wherein the first sequence (and optionally the second sequence) is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(144) 18. The vector of paragraph 17, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

(145) For example, fluorescence using the first EOU is 0.5 to X times the fluorescence using the second EOU, wherein X is from 3.0 to 1.0, eg, 3, 2.5, 2, 1.5 or 1, wherein fluorescence is determined using excitation at 481 nm and emission at 507 nm. Optionally, *E. coli* cultures at OD600 of 0.3-0.5 in the exponential growth phase are used.

(146) For example, the upstream insulator, the nucleotide sequence encoding GFP, 3' UTR, transcription terminator and downstream insulator of each EOU are as disclosed in Mutalik et al (2013). For example, the upstream insulator, the nucleotide sequence encoding GFP, 3' UTR, transcription terminator and downstream insulator of each EOU are corresponding sequences of SEQ ID NO: 4. For example, the *E. coli* is *E. coli* BW25113 is grown in MOPS EZ Rich Medium (Teknova) supplemented with 50 µg/ml kanamycin (kan) at 37° C., shaken at 900 r.p.m. For example, each EOUs is comprised by a medium copy plasmid, eg, a plasmid derived from pFAB217 comprising a p15A replication origin and a kan resistance gene.

(147) 19. The vector of any preceding paragraph, wherein the vector comprises an origin of replication that is operable in the host cell.

(148) 20. The vector of any preceding paragraph, wherein the vector comprises an origin of replication that is operable in a bacterial cell of a vector production strain, wherein the Cas3 is not operable in the production strain cell to target and cut a chromosomal sequence thereof.

(149) An example of a production strain cell is an *E. coli* cell. A production strain cell is a cell that is used to amplify DNA encoding Cas (and optionally other components of a CRISPR/Cas system). Usefully, the strain may package the amplified DNA into transduction particles that are may be isolated to produce a composition that can be contacted with a population of target host cells (eg, bacterial, archaeal, prokaryotic, eukaryotic, human, animal, mammal, rodent, mouse, rat, rabbit, Xenopus, fish, bird, amphibian, insect, plant, amoeba or algae cells) wherein the DNA is introduced into the cells for expression of the Cas (and optional other CRISPR/Cas system components), wherein the Cas is guided to a protospacer target sequence in the host cells and modifies (eg, cuts) the sequence. In another example, the amplified DNA isolated from a population of production strain cells and is combined with a delivery vehicle (eg, a carrier bacterium, nanoparticle or liposome), wherein the delivery vehicle can be contacted with a population of target host cells (eg,

bacterial, archaeal, prokaryotic, eukaryotic, human, animal, mammal, rodent, mouse, rat, rabbit, Xenopus, fish, bird, amphibian, insect, plant, amoeba or algae cells) wherein the DNA is introduced into the cells for expression of the Cas (and optional other CRISPR/Cas system components), wherein the Cas is guided to a protospacer target sequence in the host cells and modifies (eg, cuts) the sequence.

(150) 21. The vector of paragraph 20, wherein the first sequence is under the control of a promoter that is capable of controlling expression of the Cas3 at a level that is not toxic to the production strain cell.

(151) In an example, substantially no production strain cells are killed when the Cas3-encoding sequence is amplified therein. In another example, no more than 40, 30, 20, 10, 5, 4, 3, 2, or 1% of production strain cells are killed when the Cas3-encoding sequence is amplified therein. For example this is in a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 or 24 hour period of culturing the cells.

(152) 22. The vector of paragraph 20, wherein the first sequence is under the control of a promoter that controls expression of the Cas3 in the production strain cell such that the cell is capable of growth and propagation sufficient to produce at least 1000 copies of the vector.

(153) For example this is in a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 or 24 hour period of culturing the cells. For example, at least 10.sup.4, 10.sup.5, 10.sup.6, 10.sup.7, 10.sup.8, 10.sup.9, 10.sup.10, 10.sup.11, 10.sup.12, 10.sup.13, 10.sup.14, 10.sup.15, 10.sup.16, 10.sup.17 or 10.sup.18 copies of the vector are produced per 10.sup.3, 10.sup.4, 10.sup.5, 10.sup.6, 10.sup.7, 10.sup.8, 10.sup.9, 10.sup.10, 10.sup.11, 10.sup.12, 10.sup.13, 10.sup.14, 10.sup.15, 10.sup.16, 10.sup.17 production strain cells respectively.

(154) 23. The vector of any one of paragraphs 20 to 22, wherein the cell is capable of at least 2 or 3 logs of expansion when the vector is comprised therein.

(155) For example, this is in a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 or 24 hour period of culturing the cells.

(156) 24. The vector of any preceding paragraph, wherein the vector is a high copy number vector.

(157) 25. The vector of any preceding paragraph, wherein the first nucleotide sequence or operon is comprised by a mobile genetic element.

(158) Suitable mobile genetic elements, eg, transposons, are disclosed in WO2016177682 and US20170246221, the disclosures of which are explicitly incorporated herein for possible use in the invention and for providing one or more features for the claims herein.

(159) 26. The vector of any preceding paragraph, wherein the vector is devoid of a Cas adaption module.

(160) For example, the vector is devoid of nucleotide sequences encoding a Cas1, Cas2 and/or Cas4.

(161) 27. The vector of any preceding paragraph, wherein the vector is devoid of nucleotide sequence encoding one, more or all of a Cas1, Cas2, Cas4, Cas6 (optionally Cas6f), Cas7 and Cas8 (optionally Cas8f).

(162) 28. The vector of any preceding paragraph, wherein the vector is devoid of a sequence encoding a Cas6 (optionally a Cas6f).

(163) 29. The vector of any one of paragraphs 1 to 27, wherein the module encodes a Cas6 (optionally a Cas6f).

(164) 30. The vector of any preceding paragraph, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas11, Cas7 and Cas8a1.

(165) 31. The vector of paragraph 30, wherein the vector comprises nucleotide sequence encoding Cas3' and/or Cas3'' (optionally wherein the nucleotide sequences encoding the Cas3' and/or Cas3'' are between the promoter and the sequence(s) recited in paragraph 30).

(166) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3 (eg, Cas3' and/or Cas3''), Cas11, Cas7 and Cas8a1. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas11 sequence. Optionally, the vector comprises a Type IA CRISPR array or one or more nucleotide sequences encoding single

guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(167) 32. The vector of paragraph 30 or 31, wherein the host cell comprises a Type IA CRISPR array that is cognate with the Cas3.

(168) 33. The vector of paragraph 30 or 31, wherein the host cell comprises an endogenous Type IB, C, U, D, E or F CRISPR/Cas system.

(169) 34. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8b1, Cas7 and Cas5.

(170) 35. The vector of paragraph 34, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 34.

(171) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8b1, Cas7 and Cas5. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas8b1 sequence. Optionally, the vector comprises a Type IB CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(172) 36. The vector of paragraph 34 or 35, wherein the host cell comprises a Type IB CRISPR array that is cognate with the Cas3.

(173) 37. The vector of paragraph 34 or 35, wherein the host cell comprises an endogenous Type IA, C, U, D, E or F CRISPR/Cas system.

(174) 38. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas5, Cas8c and Cas7.

(175) 39. The vector of paragraph 38, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 38.

(176) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas5, Cas8c and Cas7. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas5 sequence. Optionally, the vector comprises a Type IC CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(177) 40. The vector of paragraph 38 or 39, wherein the host cell comprises a Type IC CRISPR array that is cognate with the Cas3.

(178) 41. The vector of paragraph 38 or 39, wherein the host cell comprises an endogenous Type IA, B, U, D, E or F CRISPR/Cas system.

(179) 42. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in

5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8U2, Cas7, Cas5 and Cas6.

(180) 43. The vector of paragraph 42, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 42.

(181) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8U2, Cas7, Cas5 and Cas6. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas8U2 sequence. Optionally, the vector comprises a Type IU CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(182) 44. The vector of paragraph 42 or 43, wherein the host cell comprises a Type IU CRISPR array that is cognate with the Cas3.

(183) 45. The vector of paragraph 42 or 43, wherein the host cell comprises an endogenous Type IA, B, C, D, E or F CRISPR/Cas system.

(184) 46. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas10d, Cas7 and Cas5.

(185) 47. The vector of paragraph 46, wherein the vector comprises a nucleotide sequence encoding Cas3' and/or Cas3'' (optionally wherein the nucleotide sequences encoding the Cas3' and/or Cas3'' are between the promoter and the sequence(s) recited in paragraph 46).

(186) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas10d, Cas7 and Cas5. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas10d sequence. Optionally, the vector comprises a Type ID CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(187) 48. The vector of paragraph 46 or 47, wherein the host cell comprises a Type ID CRISPR array that is cognate with the Cas3.

(188) 49. The vector of paragraph 46 or 47, wherein the host cell comprises an endogenous Type IA, B, C, U, E or F CRISPR/Cas system.

(189) 50. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8e, Cas11, Cas7, Cas5 and Cas6.

(190) 51. The vector of paragraph 50, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 50.

(191) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8e, Cas11, Cas7, Cas5 and Cas6. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas11 sequence. Optionally, the vector comprises a Type IE CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade

proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(192) 52. The vector of paragraph 50 or 51, wherein the host cell comprises a Type IE CRISPR array that is cognate with the Cas3.

(193) 53. The vector of paragraph 50 or 51, wherein the host cell comprises an endogenous Type IA, B, C, D, U or F CRISPR/Cas system.

(194) 54. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8f, Cas5, Cas7 and Cas6f.

(195) 55. The vector of paragraph 54, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 54, wherein the vector is devoid of nucleotide sequence encoding further Cas between the promoter and the sequence encoding the Cas3.

(196) In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8f, Cas5, Cas7 and Cas6f. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas8f sequence. Optionally, the vector comprises a Type IF CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

(197) 56. The vector of paragraph 54 or 55, wherein the host cell comprises a Type IF CRISPR array that is cognate with the Cas3.

(198) 57. The vector of paragraph 54 or 55, wherein the host cell comprises an endogenous Type IA, B, C, D, U or E CRISPR/Cas system.

(199) 58. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IA Cas and Cascade proteins.

(200) 59. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IB Cas and Cascade proteins.

(201) 60. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IC Cas and Cascade proteins.

(202) 61. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type ID Cas and Cascade proteins.

(203) 62. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IE Cas and Cascade proteins.

(204) 63. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IF Cas and Cascade proteins.

(205) 64. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IU Cas and Cascade proteins.

(206) 65. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *E. coli* (optionally Type IE or IF) Cas and Cascade proteins, optionally wherein the *E. coli* is ESBL-producing *E. coli* or *E. coli* ST131-O25b:H4.

(207) 66. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *Clostridium* (eg, *C. difficile*) Cas and Cascade proteins, optionally *C. difficile* resistant to one or more antibiotics selected from aminoglycosides, lincomycin, tetracyclines, erythromycin, clindamycin, penicillins, cephalosporins and fluoroquinolones.

(208) 67. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *Pseudomonas aeruginosa* Cas and Cascade proteins, optionally *P. aeruginosa* resistant to one or more antibiotics selected from carbapenems, aminoglycosides, cefepime, ceftazidime, fluoroquinolones, piperacillin and tazobactam.

(209) 68. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *Klebsiella pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*) Cas and Cascade proteins.

(210) 69. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus* or *B. halodurans* Cas and Cascade proteins.

(211) 70. The vector of any preceding paragraph, wherein the Cas3 is a Cas3 of a CRISPR/Cas locus of a first bacterial or archaeal species, wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter is further than the distance between the Cas3-encoding sequence and the respective promoter comprised by the vector.

(212) The cognate promoter here is the one that controls expression of Cas3 in the wild-type locus.

(213) 71. The vector of any preceding paragraph, wherein the distance between the promoter and the Cas3-encoding sequence and/or Cascade protein-encoding sequence(s) is shorter than in a corresponding wild-type Type I locus.

(214) A corresponding locus is a wild-type locus of a bacterial or archaeal species or strain that comprises an endogenous CRISPR/Cas system encoding the Cas3 and/or Cascade proteins of the type that are also encoded by the vector. Thus, when the vector comprises an operon, the operon may comprise Cas3- and Cascade-encoding nucleotide sequences that are not in a natural configuration.

(215) 72. The vector of any preceding paragraph, wherein the vector comprises (i) a CRISPR array for producing crRNAs in the host cell and/or (ii) one or more nucleotide sequences encoding one or more single guide RNAs (gRNAs), wherein the crRNAs or gRNAs are cognate to the Cas3 (and optionally cognate to the Cascade proteins).

(216) 73. The vector of paragraph 72 when dependent from paragraph 4, wherein the array or gRNA-encoding sequence(s) are comprised by the operon and under the control of the promoter.

(217) 74. The vector of paragraph 72, wherein the array or gRNA-encoding sequence(s) are under the control of a promoter that is different from the promoter that controls the expression of the Cas3.

(218) 75. The vector of any one of paragraphs 72 to 74, wherein one or more of the crRNAs or gRNAs comprises a spacer sequence that is capable of hybridising to a target nucleotide sequence of the host cell, wherein the target sequence is adjacent a PAM, the PAM being cognate to the Cas3.

(219) Thus, the spacer hybridises to the protospacer to guide the Cas3 to the protospacer. Optionally, the Cas3 cuts the protospacer, eg, using exo- and/or endonuclease activity of the Cas3. Optionally, the Cas3 removes a plurality (eg, at least 2, 3, 4, 5, 6, 7, 8, 9 or 10) nucleotides from the protospacer.

(220) 76. The vector of paragraph 75, wherein the target sequence is a chromosomal sequence of the host cell.

(221) 77. The vector of paragraph 75 or 76, wherein the Cas3 is operable to cut the target sequence.

(222) 78. The vector of any preceding paragraph, wherein the vector is a plasmid or phagemid.

(223) 79. A delivery vehicle comprising the vector of any preceding paragraph, wherein the delivery vehicle is capable of delivering the vector into the host cell.

(224) 80. The vehicle of paragraph 79, wherein the delivery vehicle is a phage, non-replicative transduction particle, nanoparticle carrier, bacterium or liposome.

(225) The phage or particles comprise phage coat proteins encapsidating DNA, wherein the DNA comprises the vector. Suitable examples of phage and particles are disclosed in U.S. Ser. No. 15/985,658 (and its equivalent publication by USPTO) the disclosures of which are incorporated herein by reference for possible use in the invention and for providing one or more features that

may be included in g the claims herein. Phage or particle is capable of infecting the cell, thereby introducing the vector into the cell.

(226) 81. The vector or vehicle of any preceding paragraph, wherein the host cell is a bacterial or archaeal cell, optionally, the host cell is a *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cell.

(227) 82. The vector or vehicle of any preceding paragraph for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

(228) 83. The vector or vehicle of paragraph 82, wherein the disease or condition is an infection of the subject with host cells (eg, bacterial cells), or wherein the disease or condition is mediated by host cells (eg, bacterial cells).

(229) 84. A pharmaceutical composition comprising the vector or vehicle of any preceding paragraph and a pharmaceutically acceptable diluent, excipient or carrier.

(230) 85. A method of amplifying copies of a DNA encoding a functional Cas protein (optionally a Cas nuclease) in a bacterial or archaeal production strain of cells, the method comprising (a) Providing production strain cells, each cell comprising a copy of said DNA, wherein each DNA comprises a nucleotide sequence encoding said Cas, wherein the nucleotide sequence is under the control of a promoter for controlling the expression of the Cas in the production strain cell, the DNA comprising an origin of replication that is operable in the cell for replication of the DNA; (b) Culturing the cells to allow replication of the DNA, whereby the DNA is amplified; and (c) Optionally isolating copies of the DNA,

(231) 86. The method of paragraph 85, wherein the promoter is a constitutive promoter.

(232) 87. The method of paragraph 85, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or a lac repressor).

(233) 88. The method of paragraph 85, wherein the promoter is inducible.

(234) 89. The method of any one of paragraphs 85 to 88, wherein the promoter is a medium strength promoter.

(235) 90. The method of any one of paragraphs 85 to 89, wherein the promoter has an Anderson Score (AS) of $0.5 > AS > 0.1$.

(236) 91. The method of any one of paragraphs 85 to 90, wherein the nucleotide sequence encoding said Cas is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(237) 92. The method of paragraph 91, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

(238) 93. The method of any one of paragraphs 85 to 92, wherein the nuclease is Cas3 and optionally the DNA or cell encodes cognate Cascade proteins and/or one or more crRNAs that are operable for Cas nuclease targeting.

(239) For example, the targeting is targeting of the Cas to a protospacer sequence comprised by a host cell chromosome or an episome thereof. In another example the targeting is in a recombineering method and the Cas is targeted to a protospacer sequence of a DNA that has been introduced into or amplified in the host cell. In an example of such recombineering, the host cell is an *E. coli* cell.

(240) 94. The method of any one of paragraphs 85 to 92, wherein the Cas is a Cas9.

(241) 95. The method of any one of paragraphs 85 to 92, wherein the Cas is a Type IIIA csm protein or a Type IIIB cmr protein.

(242) 96. The method of any one of paragraphs 85 to 92, wherein the Cas is a Csf1.

(243) 97. The method of any one of paragraphs 85 to 92, wherein the Cas is a Cpf1.

(244) 98. The method of any one of paragraphs 85 to 92, wherein the Cas is a Cas13 (optionally Cas13a or Cas13b).

(245) 99. The method of any one of paragraphs 85 to 92, wherein the Cas is selected from a Cas3, Cas8a, Cas5, Cas8b, Cas8c, Cas10d, Cse1, Cse2, Csy1, Csy2, Csy3, GSU0054, Cas10, Csm2, Cmr5, Cas10, Csx11, Csx10, Csf1, Cas9, Csn2, Cas4, Cpf1, C2c1, C2c3, Cas13a, Cas13b and Cas13c.

(246) 100. The method of any one of paragraphs 85 to 99, wherein the production strain cells comprise a helper phage genome that is inducible to produce phage coat proteins in the cells, wherein the method further comprises inducing production of the phage proteins and causing packaging of the amplified DNA into phage particles or non-self-replicative transduction particles, and further isolating the phage or transduction particles and optionally formulating the particles into a pharmaceutical composition for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

(247) 101. The method of paragraph 100, wherein the particles are capable of infecting target host cells in the subject and transducing the cells with the DNA, wherein the Cas and crRNAs (or gRNAs) encoded by the DNA are expressed in the cells, the crRNAs or (gRNAs) being operable to guide the Cas to a target nucleotide sequence (optionally a chromosomal sequence) comprised by the cells, wherein the Cas cuts the target sequences in the cells, thereby killing host cells and treating or reducing the risk of the disease or condition.

(248) 102. The method of paragraph 101, wherein the host cells are bacterial or archaeal cells, optionally, the host cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

(249) 103. The method of any one of paragraphs 85 to 102, wherein each DNA is comprised by a high copy number vector, optionally a high copy number plasmid (an optionally the promoter is a constitutive promoter).

(250) 104. The method of any one of paragraphs 85 to 103, wherein each DNA is comprised by a vector or vehicle according to any one of paragraphs 1 to 83.

(251) 105. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for enhancing the yield of amplified DNA produced by the production host cells.

(252) Thus, said enhancing may be relative to the yield produced using a strong promoter, eg, a strong constitutive promoter (for example a promoter having an Anderson Score (AS) of $AS \geq 0.5$). In another example, the strong promoter is a promoter comprised by a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of >4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(253) 106. The use of paragraph 105, wherein the use is for enhancing said yield by (d) reducing toxicity of the Cas in the production strain; (e) reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain; (f) promoting production cell viability during the amplification of the DNA; and/or (g) reducing the occurrence of Cas cutting of DNA (optionally cutting of production host cell chromosomal DNA or said DNA construct).

(254) 107. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing toxicity of the Cas in the production strain.

(255) 108. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain.

(256) 109. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for promoting production cell viability during the amplification of the DNA.

(257) 110. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing the occurrence of Cas cutting of DNA.

(258) 111. A method for enhancing the yield of amplified copies of a DNA construct in a population of bacterial or archaeal production strain cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

(259) 112. A method for reducing toxicity of a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

(260) 113. A method for reducing mutation of a DNA construct encoding a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

(261) 114. A method for promoting production cell viability of a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct comprised by the cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

(262) 115. A method for reducing the occurrence of Cas nuclease cutting of a DNA construct in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

(263) 116. The use of paragraph 108 or 110, or the method of paragraph 113 or 115, wherein the mutation or cutting is mutation or cutting of host cell chromosomal DNA or the construct DNA.

(264) 117. The use or method of any one of paragraphs 105 to 116, wherein the promoter is a constitutive promoter.

(265) 118. The use or method of any one of paragraphs 105 to 117, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or a lac repressor).

(266) In an example, the promoter is a constitutive promoter and optionally the DNA is comprised by a high copy number plasmid or phagemid.

(267) 119. The use or method of any one of paragraphs 105 to 118, wherein the promoter is P.sub.LtetO-1, P.sub.LlacO-1 or a repressible homologue thereof.

(268) P.sub.LlacO-1 is repressed by lac repressor (LacR). P.sub.LtetO-1 is repressed by tet repressor (TetR).

(269) 120. The use or method of any one of paragraphs 105 to 119, wherein the promoter is a medium strength promoter.

(270) 121. The use or method of any one of paragraphs 105 to 120, wherein the promoter has an Anderson Score (AS) of $0.5 > AS > 0.1$.

(271) 122. The use or method of any one of paragraphs 105 to 121, wherein the nucleotide sequence encoding said Cas is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(272) 123. The use or method of paragraph 122, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

(273) 124. The use or method of any one of paragraphs 105 to 123, wherein the nuclease is Cas3 and optionally the DNA construct encodes cognate Cascade proteins.

(274) 125. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Cas9.

(275) 126. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Type IIIA csm protein or a Type IIIB cmr protein.

(276) 127. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Csf1.

(277) 128. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Cpf1.

(278) 129. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Cas13 (optionally Cas13a or Cas13b).

(279) 130. The use or method of any one of paragraphs 105 to 123, wherein the Cas is selected from a Cas3, Cas8a, Cas5, Cas8b, Cas8c, Cas10d, Cse1, Cse2, Csy1, Csy2, Csy3, GSU0054, Cas10, Csm2, Cmr5, Cas10, Csx11, Csx10, Csf1, Cas9, Csn2, Cas4, Cpf1, C2c1, C2c3, Cas13a, Cas13b and Cas13c.

(280) 131. The use or method of any one of paragraphs 105 to 130, wherein the DNA construct comprises one or more nucleotide sequences for producing crRNAs or gRNAs that are operable for Cas nuclease targeting.

(281) 132. The use or method of any one of paragraphs 105 to 131, wherein the production strain

cells comprise a helper phage genome that is inducible to produce phage coat proteins in the cells, wherein the method further comprises inducing production of the phage proteins and causing packaging of the amplified DNA into phage particles or non-self-replicative transduction particles, and further isolating the phage or transduction particles and optionally formulating the particles into a pharmaceutical composition for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

(282) 133. The method of paragraph 132, wherein the particles are capable of infecting target host cells in the subject and transducing the cells with the DNA, wherein the Cas and crRNAs (or gRNAs) encoded by the DNA are expressed in the cells, the crRNAs or (gRNAs) being operable to guide the Cas to a target nucleotide sequence (optionally a chromosomal sequence) comprised by the cells, wherein the Cas cuts the target sequences in the cells, thereby killing host cells and treating or reducing the risk of the disease or condition.

(283) 134. The method of paragraph 133, wherein the host cells are bacterial or archaeal cells, optionally, the host cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

(284) 135. The use or method of any one of paragraphs 105 to 134, wherein each DNA is comprised by a high copy number vector, optionally a high copy number plasmid (an optionally the promoter is a constitutive promoter).

(285) 136. The use or method of any one of paragraphs 105 to 135, wherein each DNA is comprised by a vector according to any one of paragraphs 1 to 78 and 81 to 83.

(286) Clauses

(287) The invention provides, by way of example, the following Clauses; the features of these are combinable with any other disclosure herein.

(288) 1. A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

(289) 2. The vector of Clause 1, wherein the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

(290) 3. The vector of Clause 2, wherein (a) the first sequence is between the promoter and the second sequence in the operon; (b) the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence; and/or (c) the operon comprises (in 5' to 3' direction) the promoter, the first sequence and the second sequence.

(291) 4. The vector of any preceding Clause, wherein each promoter is a constitutive promoter.

(292) 5. The vector of any one of Clauses 1 to 3, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or lac repressor).

(293) 6. The vector of any one of Clauses 1 to 3, wherein the promoter is inducible.

(294) 7. The vector of any preceding Clause, wherein the first sequence is under the control of a medium strength promoter.

(295) 8. The vector of any preceding Clause, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $0.5 > AS > 0.1$.

(296) 9. The vector of any preceding Clause, wherein the first sequence (and optionally the second sequence) is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the

fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

(297) 10. The vector of Clause 9, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

(298) 11. The vector of any preceding Clause, wherein the vector comprises an origin of replication that is operable in the host cell.

(299) 12. The vector of any preceding Clause, wherein the vector comprises an origin of replication that is operable in a bacterial cell of a vector production strain, wherein the Cas3 is not operable in the production strain cell to target and cut a chromosomal sequence thereof.

(300) 13. The vector of Clause 12, wherein the first sequence is under the control of a promoter that is capable of controlling expression of the Cas3 at a level that is not toxic to the production strain cell.

(301) 14. The vector of any preceding Clause, wherein the vector is a high copy number vector.

(302) 15. The vector of any preceding Clause, wherein the first nucleotide sequence or operon is comprised by a mobile genetic element.

(303) 16. The vector of any preceding Clause, wherein the vector is devoid of a Cas adaption module.

(304) 17. The vector of any preceding Clause, wherein the vector is devoid of nucleotide sequence encoding one, more or all of a Cas1, Cas2, Cas4, Cas6, Cas7 and Cas8.

(305) 18. The vector of any preceding Clause, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas11, Cas7 and Cas8a1.

(306) 19. The vector of Clause 18, wherein the vector comprises nucleotide sequence encoding Cas3' and/or Cas3''.

(307) 20. The vector of Clause 19, wherein the nucleotide sequences encoding the Cas3' and/or Cas3'' are between the promoter and the sequence(s) recited in Clause 18.

(308) 21. The vector of any one of Clauses 18 to 20, wherein the host cell comprises a Type IA CRISPR array that is cognate with the Cas3.

(309) 22. The vector of any one of Clauses 18 to 20, wherein the host cell comprises an endogenous Type IB, C, U, D, E or F CRISPR/Cas system.

(310) 23. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8b1, Cas7 and Cas5.

(311) 24. The vector of Clause 23, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 23.

(312) 25. The vector of Clause 23 or 24, wherein the host cell comprises a Type IB CRISPR array that is cognate with the Cas3.

(313) 26. The vector of Clause 23 or 24, wherein the host cell comprises an endogenous Type IA, C, U, D, E or F CRISPR/Cas system.

(314) 27. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas5, Cas8c and Cas7.

(315) 28. The vector of Clause 27, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 27.

(316) 29. The vector of Clause 27 or 28, wherein the host cell comprises a Type IC CRISPR array that is cognate with the Cas3.

(317) 30. The vector of Clause 27 or 28, wherein the host cell comprises an endogenous Type IA, B, U, D, E or F CRISPR/Cas system.

(318) 31. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8U2, Cas7, Cas5 and Cas6.

- (319) 32. The vector of Clause 31, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 31.
- (320) 33. The vector of Clause 31 or 32, wherein the host cell comprises a Type IU CRISPR array that is cognate with the Cas3.
- (321) 34. The vector of Clause 31 or 32, wherein the host cell comprises an endogenous Type IA, B, C, D, E or F CRISPR/Cas system.
- (322) 35. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas10d, Cas7 and Cas5.
- (323) 36. The vector of Clause 35, wherein the vector comprises a nucleotide sequence encoding Cas3' and/or Cas3''.
- (324) 37. The vector of Clause 36, wherein the nucleotide sequences encoding the Cas3' and/or Cas3'' are between the promoter and the sequence(s) recited in Clause 35.
- (325) 38. The vector of any one of Clauses 35 to 37, wherein the host cell comprises a Type ID CRISPR array that is cognate with the Cas3.
- (326) 39. The vector of any one of Clauses 35 to 37, wherein the host cell comprises an endogenous Type IA, B, C, U, E or F CRISPR/Cas system.
- (327) 40. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8e, Cas11, Cas7, Cas5 and Cas6.
- (328) 41. The vector of Clause 40, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 40.
- (329) 42. The vector of Clause 40 or 41, wherein the host cell comprises a Type IE CRISPR array that is cognate with the Cas3.
- (330) 43. The vector of Clause 40 or 41, wherein the host cell comprises an endogenous Type IA, B, C, D, U or F CRISPR/Cas system.
- (331) 44. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8f, Cas5, Cas7 and Cas6f.
- (332) 45. The vector of Clause 44, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 44, wherein the vector is devoid of nucleotide sequence encoding further Cas between the promoter and the sequence encoding the Cas3.
- (333) 46. The vector of Clause 44 or 45, wherein the host cell comprises a Type IF CRISPR array that is cognate with the Cas3.
- (334) 47. The vector of Clause 44 or 45, wherein the host cell comprises an endogenous Type IA, B, C, D, U or E CRISPR/Cas system.
- (335) 48. The vector of any one of Clauses 1 to 17, wherein the Cas and Cascade are (a) Type IA Cas and Cascade proteins; (b) Type IB Cas and Cascade proteins; (c) Type IC Cas and Cascade proteins; (d) Type ID Cas and Cascade proteins; (e) Type IE Cas and Cascade proteins; (f) Type IF Cas and Cascade proteins; or (g) Type IU Cas and Cascade proteins.
- (336) 49. The vector of any preceding Clause, wherein the Cas and Cascade are *E. coli* (optionally Type IE or IF) Cas and Cascade proteins.
- (337) 50. The vector of Clause 49, wherein the *E. coli* is ESBL-producing *E. coli* or *E. coli* ST131-O25b:H4.
- (338) 51. The vector of any preceding Clause, wherein the Cas and Cascade are (a) *Clostridium* (eg, *C. difficile*) Cas and Cascade proteins, optionally *C. difficile* resistant to one or more antibiotics selected from aminoglycosides, lincomycin, tetracyclines, erythromycin, clindamycin, penicillins, cephalosporins and fluoroquinolones; (b) *Pseudomonas aeruginosa* Cas and Cascade proteins, optionally *P. aeruginosa* resistant to one or more antibiotics selected from carbapenems, aminoglycosides, cefepime, ceftazidime, fluoroquinolones, piperacillin and tazobactam; or (c) *Klebsiella pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum

Beta-Lactamase (ESBL)-producing *K. pneumoniae*) Cas and Cascade proteins.

(339) 52. The vector of any preceding Clause, wherein the Cas and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus* or *B. halodurans* Cas and Cascade proteins.

(340) 53. The vector of any preceding Clause, wherein the Cas3 is a Cas3 of a CRISPR/Cas locus of a first bacterial or archaeal species, wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter is further than the distance between the Cas3-encoding sequence and the respective promoter comprised by the vector.

(341) 54. The vector of any preceding Clause, wherein the distance between the promoter and the Cas3-encoding sequence and/or Cascade protein-encoding sequence(s) is shorter than in a corresponding wild-type Type I locus.

(342) 55. The vector of any preceding Clause, wherein the vector comprises (i) a CRISPR array for producing crRNAs in the host cell and/or (ii) one or more nucleotide sequences encoding one or more guide RNAs (gRNAs or single gRNAs), wherein the crRNAs or gRNAs are cognate to the Cas3 (and optionally cognate to the Cascade proteins).

(343) 56. The vector of Clause 55 when dependent from Clause 2, wherein the array or gRNA-encoding sequence(s) are comprised by the operon and under the control of the promoter.

(344) 57. The vector of Clause 56, wherein the array or gRNA-encoding sequence(s) are under the control of a promoter that is different from the promoter that controls the expression of the Cas3.

(345) 58. The vector of Clause 56 or 57, wherein one or more of the crRNAs or gRNAs comprises a spacer sequence that is capable of hybridising to a target nucleotide sequence of the host cell, wherein the target sequence is adjacent a PAM, the PAM being cognate to the Cas3.

(346) 59. The vector of Clause 58, wherein the target sequence is a chromosomal sequence of the host cell.

(347) 60. The vector of Clause 58 or 59, wherein the Cas3 is operable to cut the target sequence.

(348) 61. The vector of any preceding Clause, wherein the vector is a plasmid or phagemid.

(349) 62. A delivery vehicle comprising the vector of any preceding Clause, wherein the delivery vehicle is capable of delivering the vector into the host cell.

(350) 63. The vehicle of Clause 62, wherein the delivery vehicle is a phage, non-replicative transduction particle, nanoparticle carrier, bacterium or liposome.

(351) 64. The vector or vehicle of any preceding Clause, wherein the host cell is a bacterial or archaeal cell, optionally, the host cell is a *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cell.

(352) 65. The vector or vehicle of any preceding Clause for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

(353) 66. The vector or vehicle of Clause 65, wherein the disease or condition is an infection of the subject with host cells (eg, bacterial cells), or wherein the disease or condition is mediated by host cells (eg, bacterial cells).

(354) 67. A pharmaceutical composition comprising the vector or vehicle of any preceding Clause and a pharmaceutically acceptable diluent, excipient or carrier.

(355) It will be understood that particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention can be employed in various embodiments without departing from the scope of the invention. Those skilled in the art will recognize, or be able to ascertain using no more than routine study, numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims. All publications and patent applications mentioned in the specification are indicative of the level of skill of those skilled in the art to which this invention pertains. All publications and patent applications and all US equivalent patent applications and patents are herein incorporated by reference to the same extent as if each

individual publication or patent application was specifically and individually indicated to be incorporated by reference. Reference is made to WO2017/118598, US20180140698, US20170246221, US20180273940, US20160115488, US20180179547, US20170175142, US20160024510, US20150064138, US20170022499, US20160345578, US20180155729, US20180200342, WO2017112620, WO2018081502, PCT/EP2018/066954, PCT/EP2018/066980, PCT/EP2018/071454 and U.S. Ser. No. 15/985,658 and equivalent publications by the US Patent and Trademark Office (USPTO) or WIPO, the disclosures of which are incorporated herein by reference for providing disclosure that may be used in the present invention and/or to provide one or more features (eg, of a vector) that may be included in one or more claims herein.

(356) The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.” The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.” Throughout this application, the term “about” is used to indicate that a value includes the inherent variation of error for the device, the method being employed to determine the value, or the variation that exists among the study subjects.

(357) As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps

(358) The term “or combinations thereof” or similar as used herein refers to all permutations and combinations of the listed items preceding the term. For example, “A, B, C, or combinations thereof” is intended to include at least one of: A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, CBA, BCA, ACB, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB, AAA, MB, BBC, AAABCCCC, CBBAAA, CABABB, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.

(359) Any part of this disclosure may be read in combination with any other part of the disclosure, unless otherwise apparent from the context.

(360) All of the compositions and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the invention as defined by the appended claims.

(361) The present invention is described in more detail in the following non-limiting Examples.

EXAMPLES
(362) The examples illustrate fast and precision killing of *Escherichia coli* strains. As a model programmable nuclease system, we used a CRISPR guided vector (CGV™) to specifically target *Escherichia coli* MG1655.

Example 1. Single-Vector Cas3 & Cascade: Type I CRISPR-Cas System Targeting *E. coli*

(363) A plasmid (which we call a CRISPR Guided Vector™, CGV™) was constructed comprising an operon with nucleotide sequences encoding a Type I Cas3 and Cascade proteins under the control of a common promoter. *C. difficile* Type IB Cas3 and Cascade was used. A cognate CRISPR array comprising *C. difficile* repeat sequences and spacer sequence for targeting an *E. coli*

host cell chromosome was also introduced into target cells. An adaptation module containing Cas2, Cas3 and Cas4 was omitted in the vector (see FIG. 1A). In the wild-type *C. difficile* Type IB CRISPR/Cas locus, the cas3 gene is 3' of the Cascade genes (cas8b1, cas7 and cas5) and thus spaced away from the promoter upstream of the Cascade genes. When we tried this arrangement, we found killing of *E. coli* cells, but surprisingly when we changed to a synthetic operon arrangement (in 5' to 3' orientation) of promoter, cas3, cas8b1, cas7 and cas5 we saw significantly higher killing of the target *E. coli* cells.

(364) Results using this synthetic operon arrangement are shown in FIGS. 1A-1C. In FIG. 1B there is shown a dilution series (10^{sup}.1-10^{sup}.6) of drop spots (5 µl) of target *E. coli* MG1655 cells harboring the CGV on LB agar plates with and without inducers. CRISPR/Cas induction surprisingly killed 99.9% of the population (FIG. 1C, grey bar). Growth in absence of induction is shown in black. CGVTM refers to a CRISPR Guided VectorTM, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

(365) We also managed to achieve desirable targeted killing of *E. coli* cells using a similar set-up, except that *E. coli* Type IE Cas and Cascade were used, together with a cognate array targeting host cell *E. coli* chromosomal DNA (data not shown). In this case, a vector was used comprising (in 5' to 3' direction) a promoter controlling the expression of Cas3, Cas8e, Cas11, Cas7, Cas5 and Cas6 in an operon.

(366) Materials and Methods

(367) *E. coli* MG1655 was grown in lysogeny broth (LB) with shaking (250 rpm) at 37° C. When necessary, cultures were supplemented with tetracycline (10 µg/mL), and spectinomycin (400 µg/mL).

(368) To construct a plasmid containing *C. difficile* CRISPR system under arabinose inducible pBAD promoter, cas3, cas6, cas8b, cas7 and cas5 genes from *C. difficile* were amplified and cloned in a low copy number plasmid (pSC101 on). cas3 was located in the beginning of the operon followed by cas6, cas8b, cas7 and cas5. The adaptation module (consisting of cas1, cas2, and cas4) was omitted in the vector (FIG. 1A). A second plasmid containing an IPTG inducible single-spacer array targeting a chromosomal intergenic region in *E. coli* MG1655 was constructed (FIG. 1A).

The spacer was cloned under control of the IPTG-inducible P_{trc} promoter, in a CloDF13 on backbone. It contains 37 nucleotides from the genome of *E. coli* MG1655

(ctttgccgcgcgcttcgtcacgtaattctcgtcgcaa) (SEQ ID NO: 26). Additionally, the 3'-CCT protospacer adjacent motif (PAM) is located adjacent to the selected target sequence in the genome of *E. coli* MG1655 (FIG. 1A).

(369) To perform killing assays, both plasmids were transformed into *E. coli* MG1655 by electroporation. Transformants were grown in liquid LB with antibiotics to mid-log phase, and the killing efficiency was determined by serial dilution and spot plating onto LB, and LB+inducers (0.5 mM IPTG and 1 arabinose). Viability was calculated by counting colony forming units (CFUs) on the plates and data were calculated as viable cell concentration (CFU/ml).

Example 2. Single-Vector Cas3-Cascade & Array: Type I CRISPR-Cas System Targeting *E. coli*

(370) A plasmid (which we call a CRISPR Guided VectorTM, CGVTM, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components) was constructed comprising an operon with nucleotide sequences encoding a Type I Cas3 and Cascade proteins under the control of a common promoter. *C. difficile* Type IB Cas3 and Cascade was used. Adaptation module containing Cas1, Cas2 and Cas4 was omitted in the vector. A cognate CRISPR array comprising *C. difficile* repeat sequences and spacer sequence for targeting an *E. coli* host cell chromosome was also cloned in the vector (see FIG. 2A). Similarly we also constructed a plasmid comprising of an operon with nucleotide sequences encoding *E. coli* Type IE Cas3 and Cascade proteins under control of a common promoter. The *E. coli* adaption module containing Cas1 and Cas2 was omitted, in the vector. A cognate CRISPR array comprising *E. coli* repeat sequences and spacer sequence for targeting an *E. coli* host cell chromosome was also cloned in the vector.

(371) The CGV containing the *C. difficile* CRISPR-Cas system was transformed into *E. coli* MG1655 which contains a pks sequence incorporated into the genome. Results using this synthetic operon arrangement are shown in FIGS. 2A-2C. In FIG. 2B there is shown a dilution series (10^{sup}.1-10^{sup}.5) of drop spots (5 µl) of target *E. coli* MG1655 cells harboring the CGV on synthetic medium (SM) agar plates with and without inducers. CRISPR/Cas induction resulted in more than 2-log.sub.10 reductions in viable cells of *E. coli* MG1655 (FIG. 2C, grey bar). Growth in absence of induction is shown in black. CGVTM refers to a CRISPR Guided VectorTM.

(372) The survival of *E. coli* MG1655 upon induction was followed over time by plating the cultures in serial dilutions every 60 minutes, for 2 h (FIG. 3A). Killing curves revealed that CRISPR/Cas induction mediated rapid killing of *E. coli* MG1655, generating a two-log.sub.10 reduction in *E. coli* by the first 60 minutes. FIG. 3B shows a dilution series (10^{sup}.1-10^{sup}.6) of drop spots (5 µl) of induced and non-induced cultures of target *E. coli* MG1655 on SM agar plates.

(373) The CGV containing the *E. coli* CRISPR-Cas system was transformed into other *E. coli* MG1655 cells which contain a lambda sequence incorporated into the genome. Results using this synthetic operon arrangement are shown in FIGS. 6A-6B. In FIG. 6A there is shown a dilution series (10^{sup}.1-10^{sup}.5) of drop spots (5 µl) of target *E. coli* MG1655 cells harboring the CGV on synthetic medium (SM) agar plates with and without inducers. CRISPR/Cas induction resulted in more than 2-log.sub.10 reductions in viable cells of *E. coli* MG1655 (FIG. 6B, grey bar). Growth in absence of induction is shown in black. In a repeat experiment (not shown) we saw a 3-log.sub.10 reductions in viable cells of *E. coli* MG1655 with CRISPR/Cas induction.

(374) Materials and Methods

(375) *E. coli* MG1655 was grown in synthetic medium (SM) with shaking (250 rpm) at 37° C. Cultures were supplemented with 10 µg/mL tetracycline when required.

(376) To construct a plasmid containing *C. difficile* CRISPR system under arabinose inducible pBAD promoter, cas3, cas6, cas8b, cas7 and cas5 genes from *C. difficile* were amplified and cloned in a low copy number plasmid (pSC101 ori). cas3 was located in the beginning of the operon followed by cas6, cas8b, cas7 and cas5. Additionally, an IPTG inducible single-spacer array targeting a chromosomal intergenic region in *E. coli* MG1655 was included in the vector under control of the IPTG-inducible P_{trc} promoter (FIG. 2A). It contains 37 nucleotides from the PKS gene (previously integrated into the genome of *E. coli* MG1655)

(gtttggcgatggcgcggggtgtggtgtgcttcggcgt) (SEQ ID NO: 27). Additionally, the 3'-CCT protospacer adjacent motif (PAM) is located adjacent to the selected target sequence in the genome of *E. coli* MG1655 (FIG. 2A).

(377) To construct a plasmid containing *E. coli* CRISPR system under arabinose inducible pBAD promoter, cas3, cse1, cse2, cas7, cas5 and cas6 genes from *E. coli* were amplified and cloned in a low copy number plasmid (pSC101 ori). The operon comprised (in 5' to 3' direction) cas3 followed by cse1 cse2, cas7, cas5 and cas6. Additionally, an IPTG inducible single-spacer array targeting a chromosomal intergenic region in *E. coli* MG1655 was included in the vector under control of the IPTG-inducible P_{trc} promoter. It contained 32 nucleotides from the lambda sequence (previously integrated into the genome of *E. coli* MG1655) (tgggatgcctaccgcaagcagcttggcctgaa) (SEQ ID NO: 28) and found to efficiently target in Brouns et al., 2008 (Science. 2008 Aug. 15; 321(5891):960-4. doi: 10.1126/science.1159689; "Small CRISPR RNAs guide antiviral defense in prokaryotes"). Additionally, the 3'-ATG protospacer adjacent motif (PAM) is located adjacent to the selected target sequence in the genome of *E. coli* MG1655.

(378) The CGVs were transformed into *E. coli* MG1655 by electroporation. Transformants were grown in liquid SM with antibiotics to mid-log phase, and the killing efficiency was determined by serial dilution and spot plating onto LB, and LB+inducers (0.5 mM IPTG and 1% arabinose). Viability was calculated by counting colony forming units (CFUs) on the plates and data were calculated as viable cell concentration (CFU/ml).

(379) To perform killing curves, *E. coli* MG1655 harboring the CGV was grown in liquid SM with

antibiotics to mid-log phase. The culture was divided into two tubes and either inducers (0.5 mM IPTG and 1 arabinose) or PBS were added. Survival of the strain was followed over time by plating the cultures in serial dilutions (10.sup.1-10.sup.6) of drop spots (5 µl) every 60 minutes, for 2 h, on SM plates with antibiotics. Survival frequency was calculated by counting colony forming units (CFUs) on the plates and data were calculated as viable cell concentration (CFU/ml).

Example 3. Precision Killing of Target Strain *E. coli* MG1655 in a Microbiome

(380) An artificial microbial consortium was constructed to study the efficiency of the CGV carrying the CRISPR-Cas system of *C. difficile*, to specifically target *E. coli* MG1655 in the presence of other microbes, mimicking the human microbiome.

(381) The synthetic consortium consisted of three strains (two different species) with differential antibiotic resistance profiles: a streptomycin-resistant *E. coli* MG1655 (target strain), an ampicillin-resistant *E. coli* Top10, and a chloramphenicol-resistant *Lactococcus lactis* NZ9000. To create the consortium, bacterial cultures were grown separately in Brain Heart Infusion broth (BHI, optimal growth medium for *L. lactis*) to mid-log phase and mixed in fresh BHI broth with and without inducers. After 1 h induction at 30° C., the composition of the consortium was determined by counting viable colonies on selective plates. Induction of the CRISPR system in the mixed community, resulted in >10-fold killing of target *E. coli* MG1655, while leaving *E. coli* Top10 and *L. lactis* NZ9000 cell populations unharmed (FIG. 4A). In FIG. 4B there is shown a dilution series (10.sup.1-10.sup.5) of drop spots (5 µl) of the synthetic consortium after 1 h induction on BHI agar plates.

(382) Additionally, CRISPR killing of target strain *E. coli* MG1655 in the synthetic microbial consortium was compared to a pure culture (ie, target strain *E. coli* MG1655 that is not mixed with another strain or species). Unexpectedly, in both conditions, killing of 3 logs was achieved when plated on BHI agar plates with inducers (FIG. 5A). Thus, surprisingly the killing in the microbiome setting was as efficient as the killing in pure culture. In FIG. 5B there is shown a dilution series (10.sup.1-10.sup.5) of drop spots (5 µl) of the synthetic consortium and *E. coli* MG1655 in pure culture on BHI agar plates with and without inducers.

(383) Materials and Methods

(384) *E. coli* MG1655, *E. coli* Top10, and *Lactococcus lactis* NZ9000 were grown in BHI broth with shaking (250 rpm) at 30° C. Cultures were supplemented with 1000 µg/mL streptomycin, 100 µg/mL ampicillin, or 10 µg/mL chloramphenicol, respectively.

(385) To create the consortium, bacterial cultures were grown in BHI with appropriate antibiotics to mid-log phase. Cultures were washed twice in PBS to remove the antibiotics and mixed in fresh BHI broth. The mixed culture was spotted onto BHI plates with streptomycin, ampicillin or chloramphenicol to quantify the initial concentration of *E. coli* MG1655, *E. coli* Top10 and *L. lactis* NZ9000, respectively. The mixed culture was divided into two tubes and either inducers (0.5 mM IPTG and 1% arabinose) or PBS were added. After 1 h induction at 30° C., the composition of the consortium was calculated by counting colony forming units (CFUs) on selective plates and data were calculated as viable cell concentration (CFU/ml).

Example 4. Use of Promoter Repression in Vector Amplification Strains

(386) We engineered an *E. coli* Top10 production strain cell population comprising plasmid CGV DNA and an expressible sequence encoding a Tet repressor (TetR). The DNA comprised a Cas9-encoding nucleotide sequence under the control of a Tet promoter (pLtetO-1 promoter). The promoter is normally constitutively ON, but it was repressed by TetR in our cells. Thus, in this way we could successfully culture the cells and amplify the CGV without observing adverse toxicity due to Cas9 expression.

(387) In an experiment in the absence of repression, we did not observe any colonies of production strain bacteria, and we surmise that this was due to Cas9 toxicity. We believe, in addition to providing a way of increasing CGV yield (eg, for subsequent packaging into phage or non-self-replicative transduction particles), our method using repression can minimize selection for

mutations in the DNA that would otherwise be forced by higher Cas9 expression and cutting (eg, due to CGV cutting).

REFERENCES

(388) Mutalik et al, Nat Methods. 2013 April; 10(4):354-60. doi: 10.1038/nmeth. 2404. Epub 2013 Mar. 10, "Precise and reliable gene expression via standard transcription and translation initiation elements".

(389) TABLE-US-00001 TABLE 1 Example Bacteria Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table *Abiotrophia Acidocella Actinomyces Alkalilimnicola Aquaspirillum Abiotrophia defectiva Acidocella aminolytica Actinomyces bovis Alkalilimnicola ehrlichii Aquaspirillum polymorphum Acaricomes Acidocella facilis Actinomyces denticolens Alkaliphilus Aquaspirillum Acaricomes phytoseiuli Acidomonas Actinomyces europaeus Alkaliphilus oremlandii putridiconchylum Acetitomaculum Acidomonas methanolica Actinomyces georgiae Alkaliphilus transvaalensis Aquaspirillum serpens Acetitomaculum ruminis Acidothermus Actinomyces gerencseriae Allochromatium Aquimarina Acetivibrio Acidothermus cellulolyticus Actinomyces Allochromatium vinosum Aquimarina latercula Acetivibrio cellulolyticus Acidovorax hordeovulneris Alloiococcus Arcanobacterium Acetivibrio ethanolgignens Acidovorax anthurii Actinomyces howellii Alloiococcus otitis Arcanobacterium Acetivibrio multivorans Acidovorax caeni Actinomyces hyovaginalis Allokutzneria haemolyticum Acetoanaerobium Acidovorax cattleyae Actinomyces israelii Allokutzneria albata Arcanobacterium pyogenes Acetoanaerobium noterae Acidovorax citrulli Actinomyces johnsonii Altererythrobacter Archangium Acetobacter Acidovorax defluvii Actinomyces meyeri Altererythrobacter ishigakiensis Archangium gephyra Acetobacter aceti Acidovorax delafieldii Actinomyces naeslundii Altermonas Arcobacter Acetobacter cerevisiae Acidovorax facilis Actinomyces neuui Altermonas haloplanktis Arcobacter butzleri Acetobacter cibinongensis Acidovorax konjaci Actinomyces odontolyticus Altermonas macleodii Arcobacter cryaerophilus Acetobacter estunensis Acidovorax temperans Actinomyces oris Alysiella Arcobacter halophilus Acetobacter fabarum Acidovorax valerianellae Actinomyces radingae Alysiella crassa Arcobacter nitrofigilis Acetobacter ghanensis Acinetobacter Actinomyces slackii Alysiella filiformis Arcobacter skirrowii Acetobacter indonesiensis Acinetobacter baumannii Actinomyces turicensis Aminobacter Arhodomonas Acetobacter lovaniensis Acinetobacter baylyi Actinomyces viscosus Aminobacter aganoensis Arhodomonas aquaeolei Acetobacter malorum Acinetobacter bouvetii Actinoplanes Aminobacter aminovorans Arsenophonus Acetobacter nitrogenifigens Acinetobacter calcoaceticus Actinoplanes auranticolor Aminobacter niigataensis Arsenophonus nasoniae Acetobacter oeni Acinetobacter gernerii Actinoplanes brasiliensis Aminobacterium Acetobacter orientalis Acinetobacter haemolyticus Actinoplanes consettensis Aminobacterium mobile Arthrobacter Acetobacter orleanensis Acinetobacter johnsonii Actinoplanes deccanensis Aminomonas Arthrobacter agilis Acetobacter pasteurianus Acinetobacter junii Actinoplanes derwentensis Aminomonas paucivorans Arthrobacter albus Acetobacter pornorurn Acinetobacter lwoffii Actinoplanes digitatis Ammoniphilus Arthrobacter aurescens Acetobacter senegalensis Acinetobacter parvus Actinoplanes durhamensis Ammoniphilus oxalaticus Arthrobacter chlorophenolicus Acetobacter xylinus Acinetobacter radioresistens Actinoplanes ferrugineus Ammoniphilus oxalivorans Arthrobacter citreus Acetobacterium Acinetobacter schindleri Actinoplanes globisporus Amphibacillus Arthrobacter crystallopoietes Acetobacterium bakii Acinetobacter soli Actinoplanes humidus Amphibacillus xylanus Arthrobacter cummingsii Acetobacterium carbinolicum Acinetobacter tandoii Actinoplanes italicus Amphritea Arthrobacter globiformis Acetobacterium dehalogenans Acinetobacter tjernbergiae Actinoplanes liguriensis Amphritea balenae Arthrobacter Acetobacterium fimetarium Acinetobacter townneri Actinoplanes lobatus Amphritea japonica histidinovorans Acetobacterium malicum Acinetobacter ursingii Actinoplanes missouriensis Amycolatopsis Arthrobacter ilicis Acetobacterium paludosum Acinetobacter venetianus Actinoplanes palleronii Amycolatopsis alba*

Arthrobacter luteus *Acetobacterium tundrae* *Acrocarpospora Actinoplanes* *lipipinensis*
Amycolatopsis albidoflavus *Arthrobacter methylophilus* *Acetobacterium wieringae*
Acrocarpospora corrugata *Actinoplanes rectilineatus* *Amycolatopsis azurea* *Arthrobacter mysorens*
Acetobacterium woodii *Acrocarpospora Actinoplanes regularis* *Amycolatopsis coloradensis*
Arthrobacter nicotianae *Acetofilamentum macrocephala* *Actinoplanes Amycolatopsis lurida*
Arthrobacter nicotinovorans *Acetofilamentum rigidum* *Acrocarpospora pleiomorpha*
teichomyceticus *Amycolatopsis mediterranei* *Arthrobacter oxydans* *Acetohalobium Actibacter*
Actinoplanes utahensis *Amycolatopsis rifamycinica* *Arthrobacter pascens* *Acetohalobium*
arabaticum *Actibacter sediminis* *Actinopolyspora Amycolatopsis rubida* *Arthrobacter*
Acetomicrobium Actinoalloteichus Actinopolyspora halophila *Amycolatopsis sulphurea*
phenanthrenivorans *Acetomicrobium faecale* *Actinoalloteichus Actinopolyspora mortivallis*
Amycolatopsis tolypomycinica *Arthrobacter Acetomicrobium flavidum cyanogriseus Actinosynnema*
Anabaena polychromogenes *Acetonema Actinoalloteichus Actinosynnema mirum* *Anabaena*
cylindrica *Arthrobacter protophormiae* *Acetonema longum hyemiacidoni* *Actinotalea Anabaena*
flos-aquae *Arthrobacter Acetothermus Actinoalloteichus spitiensis Actinotalea fermentans*
Anabaena variabilis psychrolactophilus Acetothermus paucivorans Actinobaccillus Aerococcus
Anaeroarcus Arthrobacter ramosus Acholeplasma Actinobacillus capsulatus Aerococcus
sanguinicola Anaeroarcus burkinensis Arthrobacter sulfonivorans Acholeplasma axanthum
Actinobacillus delphinicola Aerococcus urinae Anaerobaculum Arthrobacter sulfureus
Acholeplasma brassicae Actinobacillus hominis Aerococcus urinaequi Anaerobaculum mobile
Arthrobacter uratoxydans Acholeplasma cavagenitalium Actinobacillus indolicus Aerococcus
urinaehominis Anaerobiospirillum Arthrobacter ureafaciens Acholeplasma equifetale
Actinobacillus lignieresii Aerococcus viridans Anaerobiospirillum Arthrobacter viscosus
Acholeplasma granularum Actinobacillus minor Aeromicrobium succiniciproducens Arthrobacter
woluwensis Acholeplasma hippikon Actinobacillus muris Aeromicrobium erythreum
Anaerobiospirillum thomasi *Asaia Acholeplasma laidlawii Actinobacillus Aeromonas*
Anaerococcus Asaia bogorensis Acholeplasma modicum pleuropneumoniae Aeromonas
Anaerococcus hydrogenalis Asanao Acholeplasma morum Actinobacillus porcinus
allosaccharophila Anaerococcus lactolyticus Asanao ferruginea Acholeplasma multilocale
Actinobacillus rossii Aeromonas bestiarum Anaerococcus prevotii Asticcacaulis Acholeplasma
oculi Actinobacillus scotiae Aeromonas caviae Anaerococcus tetradius Asticcacaulis
biprosthecium Acholeplasma palmarum Actinobacillus seminis Aeromonas encheleia Anaerococcus
vaginalis Asticcacaulis excentricus Acholeplasma parvum Actinobacillus succinogenes Aeromonas
Anaerofustis Atopobacter Acholeplasma pleciae Actinobacillus suis enteropelogenes Anaerofustis
stercorihominis Atopobacter phocae Acholeplasma vituli Actinobacillus ureae Aeromonas
eucrenophila Anaeromusa Atopobium Achromobacter Actinobaculum Aeromonas ichthiosmia
Anaeromusa acidaminophila Atopobium fossor Achromobacter denitrificans Actinobaculum
massiliense Aeromonas jandaei Anaeromyxobacter Atopobium minutum Achromobacter insolitus
Actinobaculum schaalii Aeromonas media Anaeromyxobacter Atopobium parvum Achromobacter
piechaudii Actinobaculum suis Aeromonas popoffii dehalogenans Atopobium rimae Achromobacter
ruhlandii Actinomyces urinale Aeromonas sobria Anaerorhabdus Atopobium vaginae
Achromobacter spanius Actinocatenispora Aeromonas veronii Anaerorhabdus furcosa
Aureobacterium Acidaminobacter Actinocatenispora rupis Agrobacterium Anaerosinus
Aureobacterium barkeri Acidaminobacter Actinocatenispora Agrobacterium Anaerosinus glycerini
Aureobacterium hydrogenoformans thailandica gelatinovorans Anaerovirgula Aureobacterium
liquefaciens Acidaminococcus Actinocatenispora sera Agrococcus Anaerovirgula multivorans
Avibacterium Acidaminococcus fermentans Actinocorallia Agrococcus citreus Ancalomicrobium
Avibacterium avium Acidaminococcus intestini Actinocorallia aurantiaca Agrococcus jenensis
Ancalomicrobium adetum Avibacterium gallinarum Acidicaldus Actinocorallia aurea Agromonas
Ancylobacter Avibacterium paragallinarum Acidicaldus organivorans Actinocorallia cavernae

Agromonas oligotrophica *Ancylobacter aquaticus* *Avibacterium volantium* *Acidimicrobium*
Actinocorallia glomerata *Agromyces Aneurinibacillus* *Azoarcus Acidimicrobium ferrooxidans*
Actinocorallia herbida *Agromyces fucosus* *Aneurinibacillus aneurinilyticus* *Azoarcus indigens*
Acidiphilium Actinocorallia libanotica *Agromyces hippuratus* *Aneurinibacillus migulanus*
Azoarcus tolulyticus *Acidiphilium acidophilum* *Actinocorallia longicatena* *Agromyces luteolus*
Aneurinibacillus Azoarcus toluvorans *Acidiphilium angustum* *Actinomadura Agromyces*
mediolanus thermoaerophilus *Azohydromonas* *Acidiphilium cryptum* *Actinomadura alba*
Agromyces ramosus *Angiococcus* *Azohydromonas australica* *Acidiphilium multivorum*
Actinomadura atramentaria *Agromyces rhizospherae* *Angiococcus disciformis* *Azohydromonas lata*
Acidiphilium organovorum *Actinomadura Akkermansia* *Angulomicrobium Azomonas* *Acidiphilium*
rubrum bangladeshensis *Akkermansia muciniphila* *Angulomicrobium tetraedrale* *Azomonas agilis*
Acidisoma Actinomadura catellatispora *Albidiferax Anoxybacillus* *Azomonas insignis* *Acidisoma*
sibiricum Actinomadura chibensis *Albidiferax ferrireducens* *Anoxybacillus pushchinoensis*
Azomonas macrocytogenes *Acidisoma tundrae* *Actinomadura chokoriensis* *Albidovulum*
Aquabacterium Azorhizobium *Acidisphaera Actinomadura citrea* *Albidovulum inexpectatum*
Aquabacterium commune *Azorhizobium caulinodans* *Acidisphaera rubrifaciens* *Actinomadura*
coerulea *Alcaligenes Aquabacterium parvum* *Azorhizophilus* *Acidithiobacillus Actinomadura*
echinospora *Alcaligenes denitrificans* *Azorhizophilus paspali* *Acidithiobacillus albertensis*
Actinomadura fibrosa *Alcaligenes faecalis* *Azospirillum* *Acidithiobacillus caldus* *Actinomadura*
formosensis *Alcanivorax Azospirillum brasilense* *Acidithiobacillus ferrooxidans* *Actinomadura*
hibisca *Alcanivorax borkumensis* *Azospirillum halopraeferens* *Acidithiobacillus thiooxidans*
Actinomadura kijaniata *Alcanivorax jadensis* *Azospirillum irakense* *Acidobacterium Actinomadura*
latina *Algicola Azotobacter* *Acidobacterium capsulatum* *Actinomadura livida* *Algicola*
bacteriolytica *Azotobacter beijerinckii* *Actinomadura Alicyclobacillus* *Azotobacter chroococcum*
luteofluorescens *Alicyclobacillus Azotobacter nigricans* *Actinomadura macra disulfidooxidans*
Azotobacter salinestris *Actinomadura madurae* *Alicyclobacillus Azotobacter vinelandii*
Actinomadura oligospora sendaiensis *Actinomadura pelletieri* *Alicyclobacillus vulcanalis*
Actinomadura rubrobrunea *Alishewanella Actinomadura rugatobispora* *Alishewanella fetalis*
Actinomadura umbrina *Alkalibacillus Actinomadura Alkalibacillus verrucosospora*
haloalkaliphilus Actinomadura vinacea *Actinomadura viridilutea* *Actinomadura viridis*
Actinomadura yumaensis *Bacillus Bacteroides Bibersteinia Borrelia Brevinema* [see below]
Bacteroides caccae *Bibersteinia trehalosi* *Borrelia afzelii* *Brevinema andersonii* *Bacteroides*
coagulans *Bifidobacterium Borrelia americana* *Brevundimonas Bacteriovorax Bacteroides*
eggerthii *Bifidobacterium adolescentis* *Borrelia burgdorferi* *Brevundimonas alba* *Bacteriovorax*
stolpii *Bacteroides fragilis* *Bifidobacterium angulatum* *Borrelia carolinensis* *Brevundimonas*
aurantiaca *Bacteroides galacturonicus* *Bifidobacterium animalis* *Borrelia coriaceae*
Brevundimonas diminuta *Bacteroides helcogenes* *Bifidobacterium asteroides* *Borrelia garinii*
Brevundimonas intermedia *Bacteroides ovatus* *Bifidobacterium bifidum* *Borrelia japonica*
Brevundimonas subvibrioides *Bacteroides pectinophilus* *Bifidobacterium boum* *Bosea*
Brevundimonas vancouverii *Bacteroides pyogenes* *Bifidobacterium breve* *Bosea minatitlanensis*
Brevundimonas variabilis *Bacteroides salyersiae* *Bifidobacterium catenulatum* *Bosea thiooxidans*
Brevundimonas vesicularis *Bacteroides stercoris* *Bifidobacterium choerinum* *Brachybacterium*
Brochothrix Bacteroides suis *Bifidobacterium coryneforme* *Brachybacterium Brochothrix*
campestris *Bacteroides tectus* *Bifidobacterium cuniculi alimentarium* *Brochothrix thermosphacta*
Bacteroides thetaiotaomicron *Bifidobacterium dentium* *Brachybacterium faecium* *Brucella*
Bacteroides uniformis *Bifidobacterium gallicum* *Brachybacterium Brucella canis* *Bacteroides*
ureolyticus *Bifidobacterium gallinarum* *paraconglomeratum* *Brucella neotomae* *Bacteroides*
vulgatus *Bifidobacterium indicum* *Brachybacterium rhamnosum* *Bryobacter Balnearium*
Bifidobacterium longum *Brachybacterium Bryobacter aggregatus* *Balnearium lithotrophicum*
Bifidobacterium tyrofermentans *Burkholderia Balneatrix magnum* *Bifidobacterium Brachyspira*

Burkholderia ambifaria Balneatrix alpica merycica Brachyspira alvinipulli Burkholderia andropogonis Balneola Bifidobacterium minimum Brachyspira hyodysenteriae Burkholderia anthina Balneola vulgaris Bifidobacterium Brachyspira innocens Burkholderia caledonica Barnesiella pseudocatenulatum Brachyspira murdochii Burkholderia caryophylli Barnesiella viscericola Bifidobacterium Brachyspira pilosicoli Burkholderia cenocepacia Bartonella pseudoIongum Burkholderia cepacia Bartonella alsatica Bifidobacterium pullorum Bradyrhizobium Burkholderia cocovenenans Bartonella bacilliformis Bifidobacterium ruminantium Bradyrhizobium canariense Burkholderia dolosa Bartonella clarridgeiae Bifidobacterium saeculare Bradyrhizobium elkanii Burkholderia fungorum Bartonella doshiae Bifidobacterium subtile Bradyrhizobium japonicum Burkholderia glathei Bartonella elizabethae Bifidobacterium Bradyrhizobium liaoningense Burkholderia glumae Bartonella grahamii thermophilum Brenneria Burkholderia graminis Bartonella henselae Bilophila Brenneria alni Burkholderia kururiensis Bartonella rochalimae Bilophila wadsworthia Brenneria nigrifluens Burkholderia multivorans Bartonella vinsonii Biostraticola Brenneria quercina Burkholderia phenazinium Bavariicoccus Biostraticola tofi Brenneria quercina Burkholderia plantarii Bavariicoccus seileri Bizionia Brenneria salicis Burkholderia pyrrocinia Bdellovibrio Bizionia argentinensis Brevibacillus Burkholderia silvatlantica Bdellovibrio bacteriovorus Blastobacter Brevibacillus agri Burkholderia stabilis Bdellovibrio exovorus Blastobacter capsulatus Brevibacillus borstelensis Burkholderia thailandensis Beggiatoa Blastobacter denitrificans Brevibacillus brevis Burkholderia tropica Beggiatoa alba Blastococcus Brevibacillus centrosporus Burkholderia unamae Beijerinckia Blastococcus aggregatus Brevibacillus choshinensis Burkholderia vietnamiensis Beijerinckia derxii Blastococcus saxobidens Brevibacillus invocatus Buttiauxella Beijerinckia fluminensis Blastochloris Brevibacillus laterosporus Buttiauxella agrestis Beijerinckia indica Blastochloris viridis Brevibacillus parabrevis Buttiauxella brennerae Beijerinckia mobilis Blastomonas Brevibacillus reuszeri Buttiauxella ferragutiae Belliella Blastomonas natatoria Brevibacterium Buttiauxella gaviniae Belliella baltica Blastopirellula Brevibacterium abidum Buttiauxella izardii Bellilinea Blastopirellula marina Brevibacterium album Buttiauxella noackiae Bellilinea caldifistulae Blautia Brevibacterium aurantiacum Buttiauxella warmboldiae Belnapia Blautia coccoides Brevibacterium celere Butyrivibrio Belnapia moabensis Blautia hansenii Brevibacterium epidermidis Butyrivibrio fibrisolvens Bergeriella Blautia producta Brevibacterium Butyrivibrio hungatei Bergeriella denitrificans Blautia wexlerae frigoritolerans Butyrivibrio proteoclasticus Beutenbergia Bogoriella Brevibacterium halotolerans Beutenbergia cavernae Bogoriella caseilytica Brevibacterium iodinum Bordetella Brevibacterium linens Bordetella avium Brevibacterium lyticum Bordetella bronchiseptica Brevibacterium mcbrellneri Bordetella hinzii Brevibacterium otitidis Bordetella holmesii Brevibacterium oxydans Bordetella parapertussis Brevibacterium paucivorans Bordetella pertussis Brevibacterium stationis Bordetella petrii Bordetella trematum Bacillus B. acidicer B. aminovorans B. glucanolyticus B. taeanensis B. lautus B. acidicola B. amylolyticus B. gordonae B. tequilensis B. lehensis B. acidiproducens B. andreesenii B. gottheilii B. thermantarcticus B. lentimorbus B. acidocaldarius B. aneurinilyticus B. graminis B. thermoaerophilus B. lentus B. acidoterrestris B. anthracis B. halmaphilus B. thermoamylovorans B. licheniformis B. aeolius B. aquimaris B. haloalkaliphilus B. thermocatenulatus B. ligniniphilus B. aerius B. arenosi B. halochares B. thermocloacae B. litoralis B. aerophilus B. arseniciselenatis B. halodenitrificans B. thermocopriae B. locisalis B. agaradhaerens B. arsenicus B. halodurans B. thermodenitrificans B. luciferensis B. agri B. aurantiacus B. halophilus B. thermoglucosidasius B. luteolus B. aidingensis B. arvi B. halosaccharovorans B. thermolactis B. luteus B. akibai B. aryabhattai B. hemicellulosilyticus B. thermoleovorans B. macauensis B. alcalophilus B. asahii B. hemicentroti B. thermophilus B. macerans B. algicola B. atrophaeus B. herbersteinensis B. thermoruber B. macquariensis B. alginolyticus B. axarquiensis B. horikoshii B. thermosphaericus B. macyae B. alkalidiazotrophicus B. azotofixans B. horneckiae B. thiaminolyticus B. malacitensis B. alkalinitrilicus B. azotoformans

B. horti *B. thioparans* *B. mannaniilyticus* *B. alkalisediminis* *B. badii* *B. huizhouensis* *B. thuringiensis* *B. marisflavi* *B. alkalitelluris* *B. barbaricus* *B. humi* *B. tianshenii* *B. marismortui* *B. altitudinis* *B. bataviensis* *B. hwajinpoensis* *B. trypoxylicola* *B. marmarensis* *B. alveayuensis* *B. beijingensis* *B. idriensis* *B. tusciae* *B. massiliensis* *B. alvei* *B. benzoovorans* *B. indicus* *B. validus* *B. megaterium* *B. amyloliquefaciens* *B. beringensis* *B. infantis* *B. vallismortis* *B. mesonae* *B. B. berkeleyi* *B. infernus* *B. vedderi* *B. methanolicus* a. subsp. *amyloliquefaciens* *B. beveridgei* *B. insolitus* *B. velezensis* *B. methylotrophicus* *B. a. subsp. plantarum* *B. bogoriensis* *B. invictae* *B. vietnamensis* *B. migulanus* *B. boroniphilus* *B. iranensis* *B. vireti* *B. mojaviensis* *B. dipsosauri* *B. borstelensis* *B. isabeliae* *B. vulcani* *B. mucilaginosus* *B. drementensis* *B. brevis* Migula *B. isronensis* *B. wakoensis* *B. muralis* *B. edaphicus* *B. butanolivorans* *B. jeotgali* *B. weihenstephanensis* *B. murimartini* *B. ehimensis* *B. canaveralius* *B. kaustophilus* *B. xiamenensis* *B. mycoides* *B. eiseniae* *B. carboniphilus* *B. kobensis* *B. xiaoxiensis* *B. naganoensis* *B. enclensis* *B. cecembensis* *B. kochii* *B. zhanjiangensis* *B. nanhaiensis* *B. endophyticus* *B. cellulossilyticus* *B. kokeshiiformis* *B. peoriae* *B. nanhaiisediminis* *B. endoradicis* *B. centrosporus* *B. koreensis* *B. persepolensis* *B. nealsonii* *B. farraginis* *B. cereus* *B. korlensis* *B. persicus* *B. neidei* *B. fastidiosus* *B. chagannorensis* *B. kribbensis* *B. pervagus* *B. neizhouensis* *B. fengqiensis* *B. chitinolyticus* *B. krulwichiae* *B. plakortidis* *B. niabensis* *B. firmus* *B. chondroitinus* *B. laevolacticus* *B. pocheonensis* *B. niacini* *B. flexus* *B. choshinensis* *B. larvae* *B. polygoni* *B. novalis* *B. foraminis* *B. chungangensis* *B. laterosporus* *B. polymyxa* *B. oceanisediminis* *B. fordii* *B. cibi* *B. salexigens* *B. popilliae* *B. odysseyi* *B. formosus* *B. circulans* *B. saliphilus* *B. pseudocalcalophilus* *B. okhensis* *B. fortis* *B. clarkii* *B. schlegelii* *B. pseudofirmus* *B. okuhidensis* *B. fumarioli* *B. clausii* *B. sediminis* *B. pseudomycoides* *B. oleronius* *B. funiculus* *B. coagulans* *B. selenatarsenatis* *B. psychrodurans* *B. oryzaecorticis* *B. fusiformis* *B. coahuilensis* *B. selenitireducens* *B. psychrophilus* *B. oshimensis* *B. galactophilus* *B. cohnii* *B. seohaeanensis* *B. psychrosaccharolyticus* *B. pabuli* *B. galactosidilyticus* *B. composti* *B. shacheensis* *B. psychrotolerans* *B. pakistanensis* *B. galliciensis* *B. curdlanolyticus* *B. shackletonii* *B. pulvifaciens* *B. pallidus* *B. gelatini* *B. cycloheptanicus* *B. siamensis* *B. pumilus* *B. pallidus* *B. gibsonii* *B. cytotoxicus* *B. silvestris* *B. purgationiresistens* *B. panacisoli* *B. ginsengi* *B. daliensis* *B. simplex* *B. pycnus* *B. panaciterrae* *B. ginsengihumi* *B. decisifrondis* *B. siralis* *B. qingdaonensis* *B. pantothenicus* *B. ginsengisoli* *B. decolorationis* *B. smithii* *B. qingshengii* *B. parabrevis* *B. globisporus* (eg, *B. B. deserti* *B. soli* *B. reuszeri* *B. paraeflexus* g. subsp. *Globisporus*; or *B. B. solimangrovi* *B. rhizosphaerae* *B. pasteurii* g. subsp. *Marinus*) *B. solisalsi* *B. rigui* *B. patagoniensis* *B. songklensis* *B. ruris* *B. sonorensis* *B. safensis* *B. sphaericus* *B. salarius* *B. sporothermodurans* *B. stearothermophilus* *B. stratosphericus* *B. subterraneus* *B. subtilis* (eg, *B. s. subsp. Inaquosorum*, or *B. s. subsp. Spizizenr*, or *B. s. subsp. Subtilis*) *Caenimonas* *Campylobacter* *Cardiobacterium* *Catenuloplanes* *Curtobacterium* *Caenimonas koreensis* *Campylobacter coli* *Cardiobacterium hominis* *Catenuloplanes atrovinosus* *Curtobacterium albidum* *Caldalkalibacillus* *Campylobacter concisus* *Carnimonas* *Catenuloplanes castaneus* *Curtobacterium citreus* *Caldalkalibacillus uzonensis* *Campylobacter curvus* *Carnimonas nigrificans* *Catenuloplanes crispus* *Caldanaerobacter* *Campylobacter fetus* *Carnobacterium* *Catenuloplanes indicus* *Caldanaerobacter subterraneus* *Campylobacter gracilis* *Carnobacterium alterfunditum* *Catenuloplanes japonicus* *Caldanaerobius* *Campylobacter helveticus* *Carnobacterium divergens* *Catenuloplanes nepalensis* *Caldanaerobius fijiensis* *Campylobacter hominis* *Carnobacterium funditum* *Catenuloplanes niger* *Caldanaerobius* *Campylobacter hyointestinalis* *Carnobacterium gallinarum* *Chryseobacterium polysaccharolyticus* *Campylobacter jejuni* *Carnobacterium* *Chryseobacterium* *Caldanaerobius zeae* *Campylobacter lari* *maltaromaticum* *balustinum* *Caldanaerovirga* *Campylobacter mucosalis* *Carnobacterium mobile* *Citrobacter* *Caldanaerovirga acetigignens* *Campylobacter rectus* *Carnobacterium viridans* *C. amalonaticus* *Caldicellulosiruptor* *Campylobacter showae* *Caryophanon* *C. braakii* *Caldicellulosiruptor bescii* *Campylobacter sputorum* *Caryophanon latum* *C. diversus* *Caldicellulosiruptor kristjanssonii* *Campylobacter upsaliensis* *Caryophanon tenue* *C. farmeri* *Caldicellulosiruptor owensensis* *Capnocytophaga* *Catellatospora* *C. freundii*

Capnocytophaga canimorsus *Catellatospora citrea* *C. gillenbergii* *Capnocytophaga cynodegmi*
Catellatospora *C. koseri* *Capnocytophaga gingivalis* *methionotrophica* *C. murlinae*
Capnocytophaga granulosa *Catenococcus* *C. pasteurii*.sup.[1] *Capnocytophaga haemolytica*
Catenococcus thiocycli *C. rodentium* *Capnocytophaga ochracea* *C. sedlakii* *Capnocytophaga*
sputigena *C. werkmanii* *C. youngae* *Clostridium* (see below) *Coccochloris* *Coccochloris elabens*
Corynebacterium *Corynebacterium flavescens* *Corynebacterium variabile* *Clostridium* *Clostridium*
absonum, *Clostridium aceticum*, *Clostridium acetireducens*, *Clostridium acetobutylicum*,
Clostridium acidisoli, *Clostridium aciditolerans*, *Clostridium acidurici*, *Clostridium aerotolerans*,
Clostridium aestuarii, *Clostridium akagii*, *Clostridium aldenense*, *Clostridium aldrichii*,
Clostridium algidicarni, *Clostridium algidixylanolyticum*, *Clostridium algifaecis*, *Clostridium*
algoriphilum, *Clostridium alkalicellulosi*, *Clostridium aminophilum*, *Clostridium aminovalericum*,
Clostridium amygdalinum, *Clostridium amylolyticum*, *Clostridium arbusti*, *Clostridium arcticum*,
Clostridium argentinense, *Clostridium asparagiforme*, *Clostridium aurantibutyricum*, *Clostridium*
autoethanogenum, *Clostridium baratii*, *Clostridium barkeri*, *Clostridium bartlettii*, *Clostridium*
beijerinckii, *Clostridium bifermentans*, *Clostridium bolteae*, *Clostridium bornimense*, *Clostridium*
botulinum, *Clostridium bowmanii*, *Clostridium bryantii*, *Clostridium butyricum*, *Clostridium*
cadaveris, *Clostridium caenicola*, *Clostridium caminithermale*, *Clostridium carboxidivorans*,
Clostridium carnis, *Clostridium cavendishii*, *Clostridium celatum*, *Clostridium celerecrescens*,
Clostridium cellobioparum, *Clostridium cellulofragmentans*, *Clostridium cellulolyticum*,
Clostridium cellulosi, *Clostridium cellulovorans*, *Clostridium chartatabidum*, *Clostridium*
chauvoei, *Clostridium chromiireducens*, *Clostridium citroniae*, *Clostridium clariflavum*,
Clostridium clostridioforme, *Clostridium coccoides*, *Clostridium cochlearium*, *Clostridium*
colletant, *Clostridium colicanis*, *Clostridium colinum*, *Clostridium collagenovorans*, *Clostridium*
cylindrosporum, *Clostridium difficile*, *Clostridium diolis*, *Clostridium disporicum*, *Clostridium*
drakei, *Clostridium durum*, *Clostridium estertheticum*, *Clostridium estertheticum estertheticum*,
Clostridium estertheticum laramiense, *Clostridium fallax*, *Clostridium felsineum*, *Clostridium*
fervidum, *Clostridium fimetarium*, *Clostridium formicaceticum*, *Clostridium frigidicarnis*,
Clostridium frigoris, *Clostridium ganghwense*, *Clostridium gasigenes*, *Clostridium ghonii*,
Clostridium glycolicum, *Clostridium glycyrrhizinilyticum*, *Clostridium grantii*, *Clostridium*
haemolyticum, *Clostridium halophilum*, *Clostridium hastiforme*, *Clostridium hathewayi*,
Clostridium herbivorans, *Clostridium hiranonis*, *Clostridium histolyticum*, *Clostridium*
homopropionicum, *Clostridium huakuii*, *Clostridium hungatei*, *Clostridium hydrogeniformans*,
Clostridium hydroxybenzoicum, *Clostridium hylemonae*, *Clostridium jejuense*, *Clostridium indolis*,
Clostridium innocuum, *Clostridium intestinale*, *Clostridium irregulare*, *Clostridium isatidis*,
Clostridium josui, *Clostridium kluyveri*, *Clostridium lactatifermentans*, *Clostridium*
lacusfryxellense, *Clostridium laramiense*, *Clostridium lavalense*, *Clostridium lentocellum*,
Clostridium lentoputrescens, *Clostridium leptum*, *Clostridium limosum*, *Clostridium litorale*,
Clostridium lituseburense, *Clostridium ljungdahlii*, *Clostridium lortetii*, *Clostridium lundense*,
Clostridium magnum, *Clostridium malenominatum*, *Clostridium manganotii*, *Clostridium*
mayombeii, *Clostridium methoxybenzovorans*, *Clostridium methylpentosum*, *Clostridium*
neopropionicum, *Clostridium nexile*, *Clostridium nitrophenolicum*, *Clostridium novyi*, *Clostridium*
oceanicum, *Clostridium orbiscindens*, *Clostridium oroticum*, *Clostridium oxalicum*, *Clostridium*
papyrosolvans, *Clostridium paradoxum*, *Clostridium paraperfringens* (Alias: *C. welchii*),
Clostridium paraputrificum, *Clostridium pascui*, *Clostridium pasteurianum*, *Clostridium*
peptidivorans, *Clostridium perenne*, *Clostridium perfringens*, *Clostridium pfennigii*, *Clostridium*
phytofermentans, *Clostridium piliforme*, *Clostridium polysaccharolyticum*, *Clostridium populeti*,
Clostridium propionicum, *Clostridium proteoclasticum*, *Clostridium proteolyticum*, *Clostridium*
psychrophilum, *Clostridium puniceum*, *Clostridium purinilyticum*, *Clostridium putrefaciens*,
Clostridium putrificum, *Clostridium quercicolum*, *Clostridium quinii*, *Clostridium ramosum*,
Clostridium rectum, *Clostridium roseum*, *Clostridium saccharobutylicum*, *Clostridium*

saccharogumia, *Clostridium saccharolyticum*, *Clostridium saccharoperbutylacetonicum*, *Clostridium sardiniense*, *Clostridium sartagoforme*, *Clostridium scatologenes*, *Clostridium schirmacherense*, *Clostridium scindens*, *Clostridium septicum*, *Clostridium sordellii*, *Clostridium sphenoides*, *Clostridium spiroforme*, *Clostridium sporogenes*, *Clostridium sporosphaeroides*, *Clostridium stercorarium*, *Clostridium stercorarium leptospartum*, *Clostridium stercorarium stercorarium*, *Clostridium stercorarium thermolacticum*, *Clostridium sticklandii*, *Clostridium straminisolvens*, *Clostridium subterminale*, *Clostridium sufflavum*, *Clostridium sulfidigenes*, *Clostridium symbiosum*, *Clostridium tagluense*, *Clostridium tepidiprofundum*, *Clostridium termitidis*, *Clostridium tertium*, *Clostridium tetani*, *Clostridium tetanomorphum*, *Clostridium thermaceticum*, *Clostridium thermautotrophicum*, *Clostridium thermoalcaliphilum*, *Clostridium thermobutyricum*, *Clostridium thermocellum*, *Clostridium thermocopriae*, *Clostridium thermohydrosulfuricum*, *Clostridium thermolacticum*, *Clostridium thermopalmarium*, *Clostridium thermopapyrolyticum*, *Clostridium thermosaccharolyticum*, *Clostridium thermosuccinogenes*, *Clostridium thermosulfurigenes*, *Clostridium thiosulfatireducens*, *Clostridium tyrobutyricum*, *Clostridium uliginosum*, *Clostridium ultunense*, *Clostridium villosum*, *Clostridium vincentii*, *Clostridium viride*, *Clostridium xylanolyticum*, *Clostridium xylanovorans* *Dactylosporangium* *Deinococcus* *Delftia* *Echinicola* *Dactylosporangium aurantiacum* *Deinococcus aerius* *Delftia acidovorans* *Echinicola pacifica* *Dactylosporangium fulvum* *Deinococcus apachensis* *Desulfovibrio* *Echinicola vietnamensis* *Dactylosporangium matsuzakiense* *Deinococcus aquaticus* *Desulfovibrio desulfuricans* *Dactylosporangium roseum* *Deinococcus aquatilis* *Diplococcus* *Dactylosporangium thailandense* *Deinococcus caeni* *Diplococcus pneumoniae* *Dactylosporangium vinaceum* *Deinococcus radiodurans* *Deinococcus radiophilus* *Enterobacter* *Enterobacter kobei* *Faecalibacterium* *Flavobacterium* *E. aerogenes* *E. ludwigii* *Faecalibacterium prausnitzii* *Flavobacterium antarcticum* *E. amnigemis* *E. mori* *Fangia* *Flavobacterium aquatile* *E. agglomerans* *E. nimipressuralis* *Fangia hongkongensis* *Flavobacterium aquidurens* *E. arachidis* *E. oryzae* *Fastidiosipila* *Flavobacterium balustinum* *E. asburiae* *E. pulveris* *Fastidiosipila sanguinis* *Flavobacterium croceum* *E. cancerogenus* *E. pyrinus* *Fusobacterium* *Flavobacterium cucumis* *E. cloacae* *E. radicincitans* *Fusobacterium nucleatum* *Flavobacterium daejeonense* *E. cowanii* *E. taylorae* *Flavobacterium defluvii* *E. dissolvens* *E. turicensis* *Flavobacterium degerlachei* *E. gergoviae* *E. sakazakii* *Enterobacter soli* *Flavobacterium* *E. helveticus* *Enterococcus denitrificans* *E. hormaechei* *Enterococcus durans* *Flavobacterium filum* *E. intermedins* *Enterococcus faecalis* *Flavobacterium flevense* *Enterococcus faecium* *Flavobacterium frigidarium* *Erwinia* *Flavobacterium mizutaii* *Erwinia hapontici* *Flavobacterium* *Escherichia okeanokoites* *Escherichia coli* *Gaetbulibacter* *Haemophilus* *Ideonella* *Janibacter* *Gaetbulibacter saemankumensis* *Elaemophilus aegyptius* *Ideonella azotifigens* *Janibacter anophelis* *Gallibacterium* *Elaemophilus aphrophilus* *Idiomarina* *Janibacter corallicola* *Gallibacterium anatis* *Haemophilus felis* *Idiomarina abyssalis* *Janibacter limosus* *Gallicola* *Haemophilus gallinarum* *Idiomarina baltica* *Janibacter melonis* *Gallicola barnesae* *Haemophilus haemolyticus* *Idiomarina fontislapidosi* *Janibacter terrae* *Garciella* *Haemophilus influenzae* *Idiomarina loihiensis* *Jannaschia* *Garciella nitratreducens* *Haemophilus paracuniculus* *Idiomarina ramblicola* *Jannaschia cystaugens* *Geobacillus* *Haemophilus parahaemolyticus* *Idiomarina seosinensis* *Jannaschia helgolandensis* *Geobacillus thermoglucosidasius* *Haemophilus parainfluenzae* *Idiomarina zobellii* *Jannaschia pohangensis* *Geobacillus stearothermophilus* *Haemophilus Ignatzschineria* *Jannaschia rubra* *Geobacter paraphrohaemolyticus* *Ignatzschineria larvae* *Geobacter bemidjiensis* *Haemophilus parasuis* *Janthinobacterium* *Geobacter bremensis* *Haemophilus pittmaniae* *Ignavigranum* *Janthinobacterium* *Geobacter chapellei* *Hafnia* *Ignavigranum ruoffiae* *agaricidamnosum* *Geobacter grbiciae* *Hafnia alvei* *Ilumatobacter* *Janthinobacterium lividum* *Geobacter hydrogenophilus* *Hahella* *Ilumatobacter fluminis* *Jejuia* *Geobacter lovleyi* *Hahella ganghwensis* *Ilyobacter* *Jejuia pallidilutea* *Geobacter metallireducens* *Halalkalibacillus* *Ilyobacter delafieldii* *Jeotgalibacillus* *Geobacter pelophilus* *Halalkalibacillus*

halophilus Ilyobacter insuetus Jeotgalibacillus Geobacter pickeringii Helicobacter Ilyobacter
polytropus alimentarius Geobacter sulfurreducens Helicobacter pylori Ilyobacter tartaricus
Jeotgalicoccus Geodermatophilus Jeotgalicoccus halotolerans Geodermatophilus obscurus
Gluconacetobacter Gluconacetobacter xylinus Gordonia Gordonia rubripertincta Kaistia
Labeledella Listeria ivanovii Micrococcus Nesterenkonia Kaistia adipata Labeledella gwakjiensis L.
marthii Micrococcus luteus Nesterenkonia holobia Kaistia soli Labrenzia L. monocytogenes
Micrococcus lylae Nocardia Kangiella Labrenzia aggregata L. newyorkensis Moraxella Nocardia
argentinensis Kangiella aquimarina Labrenzia alba L. riparia Moraxella bovis Nocardia corallina
Kangiella koreensis Labrenzia alexandrii L. rocourtiae Moraxella nonliquefaciens Nocardia
Labrenzia marina L. seeligeri Moraxella osloensis otitidiscaviarum Kerstersia Labrys L.
weihenstephanensis Nakamurella Kerstersia gyiorum Labrys methylaminiphilus L. welshimeri
Nakamurella multipartita Kiloniella Labrys miyagiensis Listonella Nannocystis Kiloniella
laminariae Labrys monachus Listonella anguillarum Nannocystis pusilia Klebsiella Labrys
okinawensis Macrococcus Natranaerobius K. gramilomatis Labrys portucalensis Macrococcus
bovicus Natranaerobius K. oxytoca Marinobacter thermophilus K. pneumoniae Lactobacillus
Marinobacter algicola Natranaerobius trueperi K. terrigena [see below] Marinobacter
bryozorum Naxibacter K. variicola Laceyella Marinobacter flavimaris Naxibacter alkalitolerans
Kluyvera Laceyella putida Meiothermus Neisseria Kluyvera ascorbata Lechevalieria Meiothermus
ruber Neisseria cinerea Kocuria Lechevalieria aerocolonigenes Methylophilus Neisseria
denitrificans Kocuria roasea Legionella Methylophilus methylotrophus Neisseria gonorrhoeae
Kocuria varians [see below] Microbacterium Neisseria lactamica Kurthia Listeria Microbacterium
Neisseria mucosa Kurthia zopfii L. aquatica ammoniaphilum Neisseria sicca L. booriae
Microbacterium arborescens Neisseria subflava L. cornellensis Microbacterium liquefaciens
Neptunomonas L. fleischmannii Microbacterium oxydans Neptunomonas japonica L. floridensis L.
grandensis L. grayi L. innocua Lactobacillus L. acetotolerans L. cateniformis L. mali L. parakefiri
L. sakei L. acidifarinae L. ceti L. manihotivorans L. paralimentarius L. salivarius L. acidipiscis L.
coleohominis L. mindensis L. paraplantarum L. sanfranciscensis L. acidophilus L. collinoides L.
mucosae L. pentosus L. satsumensis Lactobacillus agilis L. composti L. murinus L. perolens L.
secaliphilus L. algidus L. concavus L. nagelii L. plantarum L. sharpeae L. alimentarius L.
coryniformis L. namurensis L. pontis L. siliginis L. amylolyticus L. crispatus L. nantensis L.
protectus L. spicheri L. amylophilus L. crustorum L. oligofermentans L. psittaci L. suebicus L.
amylotrophicus L. curvatus L. oris L. rennini L. thailandensis L. amylovorus L. delbrueckii subsp.
bulgaricus L. panis L. reuteri L. ultunensis L. animalis L. delbrueckii subsp. L. pantheris L.
rhamnosus L. vaccinostercus L. antri delbrueckii L. parabrevis L. rimae L. vaginalis L. apodemi L.
delbrueckii subsp. lactis L. parabuchneri L. rogosae L. versmoldensis L. aviarius L. dextrinicus L.
paracasei L. rossiae L. vini L. bifermentans L. diolivorans L. paracollinoides L. ruminis L.
vitulinus L. brevis L. equi L. parafarraginis L. saerimneri L. zae L. buchneri L. equigenerosi L.
homohiochii L. jensenii L. zymae L. camelliae L. farraginis L. iners L. johnsonii L. gastricus L.
casei L. farciminis L. ingluviei L. kalixensis L. ghanensis L. kitasatonis L. fermentum L. intestinalis
L. kefiranofermentans L. graminis L. kunkei L. fornicalis L. fuchuensis L. kefiri L. hammesii L.
leichmannii L. fructivorans L. gallinarum L. kimchii L. hamsteri L. lindneri L. frumenti L. gasseri
L. helveticus L. harbinensis L. malefermentans L. hilgardii L. hayakitensis Legionella Legionella
adelaidensis Legionella drancourtii Candidatus Legionella jeonii Legionella quinlivanii Legionella
anisa Legionella dresdenensis Legionella jordanis Legionella rowbothamii Legionella beliardensis
Legionella drozanskii Legionella lansingensis Legionella rubrilucens Legionella birminghamensis
Legionella dumoffii Legionella londiniensis Legionella sainthelensi Legionella bozemanae
Legionella erythra Legionella longbeachae Legionella santicrucis Legionella brunensis Legionella
fairfieldensis Legionella lytica Legionella shakespearei Legionella busanensis Legionella fallonii
Legionella maceachernii Legionella spiritensis Legionella cardiaca Legionella feeleei Legionella
massiliensis Legionella steelei Legionella cherrii Legionella geestiana Legionella micdadei

Legionella steigerwaltii *Legionella cincinnatiensis* *Legionella genomospecies* *Legionella monrovia* *Legionella taurinensis* *Legionella clemsonensis* *Legionella gormanii* *Legionella moravica* *Legionella tucsonensis* *Legionella donaldsonii* *Legionella gratiana* *Legionella nagasakiensis* *Legionella tunisiensis* *Legionella gresilensis* *Legionella nautarum* *Legionella wadsworthii* *Legionella hackeliae* *Legionella norrlandica* *Legionella waltersii* *Legionella impletisoli* *Legionella oakridgensis* *Legionella worsleiensis* *Legionella israelensis* *Legionella parisiensis* *Legionella yabuuchiae* *Legionella jamestowniensis* *Legionella pittsburghensis* *Legionella pneumophila* *Legionella quateirensis* *Oceanibulbus* *Paenibacillus* *Prevotella* *Quadrisphaera* *Oceanibulbus indolifex* *Paenibacillus thiaminolyticus* *Prevotella albensis* *Quadrisphaera granulorum* *Oceanicaulis* *Pantoea* *Prevotella amnii* *Quatrionicoccus* *Oceanicaulis alexandrii* *Pantoea agglomerans* *Prevotella bergensis* *Quatrionicoccus* *Oceanicola* *Prevotella bivia australiensis* *Oceanicola batsensis* *Paracoccus* *Prevotella brevis* *Oceanicola granulosus* *Paracoccus alcaliphilus* *Prevotella bryantii* *Quinella* *Oceanicola nanhaiensis* *Paucimonas* *Prevotella buccae* *Quinella ovalis* *Oceanimonas* *Paucimonas lemoignei* *Prevotella buccalis* *Oceanimonas baumannii* *Pectobacterium* *Prevotella copri* *Ralstonia* *Oceaniserpentilla* *Pectobacterium aroidearum* *Prevotella dentalis* *Ralstonia eutropha* *Oceaniserpentilla haliotis* *Pectobacterium atrosepticum* *Prevotella denticola* *Ralstonia insidiosa* *Oceanisphaera* *Pectobacterium betavascularum* *Prevotella disiens* *Ralstonia mannitolilytica* *Oceanisphaera donghaensis* *Pectobacterium cacticida* *Prevotella histicola* *Ralstonia pickettii* *Oceanisphaera litoralis* *Pectobacterium carnegieana* *Prevotella intermedia* *Ralstonia* *Oceanithermus* *Pectobacterium carotovorum* *Prevotella maculosa* *pseudosolanacearum* *Oceanithermus desulfurans* *Pectobacterium chrysanthemi* *Prevotella marshii* *Ralstonia syzygii* *Oceanithermus profundus* *Pectobacterium cypripedii* *Prevotella melaninogenica* *Ralstonia solanacearum* *Oceanobacillus* *Pectobacterium rhapontici* *Prevotella micans* *Ramlibacter* *Oceanobacillus caeni* *Pectobacterium wasabiae* *Prevotella multiformis* *Ramlibacter henchirensis* *Oceanospirillum* *Pianococcus* *Prevotella nigrescens* *Ramlibacter tataouinensis* *Oceanospirillum linum* *Pianococcus citreus* *Prevotella oralis* *Planomicrobium* *Prevotella oris* *Raoultella* *Planomicrobium okeanoikoites* *Prevotella oulorum* *Raoultella ornithinolytica* *Plesiomonas* *Prevotella pallens* *Raoultella planticola* *Plesiomonas shigelloides* *Prevotella salivae* *Raoultella terrigena* *Proteus* *Prevotella stercorea* *Rathayibacter* *Proteus vulgaris* *Prevotella tanneriae* *Rathayibacter caricis* *Prevotella timonensis* *Rathayibacter festucae* *Prevotella veroralis* *Rathayibacter iranicus* *Providencia* *Rathayibacter rathayi* *Providencia stuartii* *Rathayibacter toxicus* *Pseudomonas* *Rathayibacter tritici* *Pseudomonas aeruginosa* *Rhodobacter* *Pseudomonas alcaligenes* *Rhodobacter sphaeroides* *Pseudomonas anguillispetica* *Ruegeria* *Pseudomonas fluorescens* *Ruegeria gelatinovorans* *Pseudoalteromonas haloplanktis* *Pseudomonas mendocina* *Pseudomonas pseudoalcaligenes* *Pseudomonas putida* *Pseudomonas tutzeri* *Pseudomonas syringae* *Psychrobacter* *Psychrobacter faecalis* *Psychrobacter phenylpyruvicus* *Saccharococcus* *Sagittula* *Sanguibacter* *Stenotrophomonas* *Tatlockia* *Saccharococcus thermophilus* *Sagittula stellata* *Sanguibacter keddiei* *Stenotrophomonas* *Tatlockia maceachernii* *Saccharomonospora* *Salegentibacter* *Sanguibacter suarezii* *maltophilia* *Tatlockia micdadei* *Saccharomonospora azurea* *Salegentibacter salegens* *Saprospira* *Streptococcus* *Tenacibaculum* *Saccharomonospora cyanea* *Salimicrobium* *Saprospira grandis* *Tenacibaculum* *Saccharomonospora viridis* *Salimicrobium album* *Sarcina* [also see below] *amylolyticum* *Saccharophagus* *Salinibacter* *Sarcina maxima* *Streptomyces* *Tenacibaculum discolor* *Saccharophagus degradans* *Salinibacter ruber* *Sarcina ventriculi* *Streptomyces* *Tenacibaculum* *Saccharopolyspora* *Salinicoccus* *Sebaldella achromogenes* *gallaicum* *Saccharopolyspora erythraea* *Salinicoccus alkaliphilus* *Sebaldella termitidis* *Streptomyces cesalbus* *Tenacibaculum* *Saccharopolyspora gregorii* *Salinicoccus hispanicus* *Streptomyces cescaepitosus* *lutimaris* *Saccharopolyspora hirsuta* *Salinicoccus roseus* *Serratia* *Streptomyces cesdiastaticus* *Tenacibaculum* *Saccharopolyspora hordei* *Salinispora* *Serratia fonticola* *Streptomyces cesexfoliatus* *mesophilum* *Saccharopolyspora rectivirgula* *Salinispora arenicola* *Serratia*

marcescens Streptomyces fimbriatus Tenacibaculum Saccharopolyspora spinosa Salinispora tropica Sphaerotilus Streptomyces fradiae skagerrakense Saccharopolyspora taberi Salinivibrio Sphaerotilus natans Streptomyces fulvissimus Tepidanaerobacter Saccharothrix Salinivibrio costicola Sphingobacterium Streptomyces griseoruber Tepidanaerobacter Saccharothrix australiensis Salmonella Sphingobacterium multivorum Streptomyces griseus syntrophicus Saccharothrix coeruleofusca Salmonella bongori Staphylococcus Streptomyces lavendulae Tepidibacter Saccharothrix espanaensis Salmonella enterica [see below] Streptomyces Tepidibacter Saccharothrix longispora Salmonella subterranea phaeochromogenes formicigenes Saccharothrix mutabilis Salmonella typhi Streptomyces Tepidibacter thalassicus Saccharothrix syringae thermodiastaticus Thermus Saccharothrix tangerinus Streptomyces tubercidicus Thermus aquaticus Saccharothrix texasensis Thermus filiformis Thermus thermophilus Staphylococcus S. arlettae S. equorum S. microti S. schleiferi S. agnetis S. felis S. muscae S. sciuri S. aureus S. fleurettii S. nepalensis S. simiae S. auricularis S. gallinarum S. pasteurii S. simulans S. capitis S. haemolyticus S. petrasii S. stepanovicii S. caprae S. hominis S. pettenkoferi S. succinus S. carnosus S. hyicus S. piscifermentans S. vitulinus S. caseolyticus S. intermedius S. pseudintermedius S. warneri S. chromogenes S. kloosii S. pseudolugdunensis S. xylosus S. cohnii S. leei S. pulvereri S. condimenti S. lentus S. rostri S. delphini S. lugdunensis S. saccharolyticus S. devriesei S. lutrae S. saprophyticus S. epidermidis S. lyticans S. massiliensis Streptococcus Streptococcus agalactiae Streptococcus infantarius Streptococcus orisratti Streptococcus thermophilus Streptococcus anginosus Streptococcus iniae Streptococcus parasanguinis Streptococcus sanguinis Streptococcus bovis Streptococcus intermedius Streptococcus peroris Streptococcus sobrinus Streptococcus canis Streptococcus lactarius Streptococcus pneumoniae Streptococcus suis Streptococcus constellatus Streptococcus milleri Streptococcus Streptococcus uberis Streptococcus downei Streptococcus mitis Streptococcus pseudopneumoniae Streptococcus vestibularis Streptococcus dysgalactiae Streptococcus mutans Streptococcus pyogenes Streptococcus viridans Streptococcus equines Streptococcus oralis Streptococcus ratti Streptococcus Streptococcus faecalis Streptococcus tigurinus Streptococcus salivarius zooepidemicus Streptococcus ferus Uliginosibacterium Vagococcus Vibrio Virgibacillus Xanthobacter Vagococcus carniphilus Vibrio aerogenes Virgibacillus Xanthobacter agilis Uliginosibacterium gangwonense Vagococcus elongatus Vibrio aestuarianus halodenitrificans Xanthobacter Ulvibacter Vagococcus fessus Vibrio albensis Virgibacillus aminoxidans Ulvibacter litoralis Vagococcus fluvialis Vibrio alginolyticus pantothenicus Xanthobacter Umezawaea Vagococcus lutrae Vibrio campbellii Weissella autotrophicus Umezawaea tangerina Vagococcus salmoninarum Vibrio cholerae Weissella cibaria Xanthobacter flavus Undibacterium Variovorax Vibrio cincinnatiensis Weissella confusa Xanthobacter tagetidis Undibacterium pigrum Variovorax boronicumulans Vibrio coralliilyticus Weissella halotolerans Xanthobacter viscosus Ureaplasma Variovorax dokdonensis Vibrio cyclitrophicus Weissella hellenica Xanthomonas Ureaplasma urealyticum Variovorax paradoxus Vibrio diazotrophicus Weissella kandleri Xanthomonas Variovorax soli Vibrio fluvialis Weissella koreensis albilineans Ureibacillus Veillonella Vibrio furnissii Weissella minor Xanthomonas alfalfae Ureibacillus composti Veillonella atypica Vibrio gazogenes Weissella Xanthomonas Ureibacillus suwonensis Veillonella caviae Vibrio haliotocoli paramesenteroides arboricola Ureibacillus terrenus Veillonella criceti Vibrio harveyi Weissella soli Xanthomonas Ureibacillus thermophilus Veillonella dispar Vibrio ichthyenteri Weissella thailandensis axonopodis Ureibacillus thermosphaericus Veillonella montpellierensis Vibrio mediterranei Weissella viridescens Xanthomonas Veillonella parvula Vibrio metschnikovii Williamsia campestris Veillonella ratti Vibrio mytili Williamsia marianensis Xanthomonas citri Veillonella rodentium Vibrio natriegens Williamsia maris Xanthomonas codiae Venenivibrio Vibrio navarrensis Williamsia serinedens Xanthomonas Venenivibrio stagnispumantis Vibrio nereis Winogradskyella cucurbitae Vibrio nigripulchritudo Winogradskyella Xanthomonas Verminephrobacter Vibrio ordalii thalassocola euvesicatoria Verminephrobacter eiseniae Vibrio orientalis Wolbachia Xanthomonas fragariae Vibrio parahaemolyticus Wolbachia persica

Xanthomonas fuscans *Verrucomicrobium* *Vibrio pectenicida* *Xanthomonas gardneri*
Verrucomicrobium spinosum *Vibrio penaeicida* *Wolinella* *Xanthomonas hortorum* *Vibrio*
proteolyticus *Wolinella succinogenes* *Xanthomonas hyacinthi* *Vibrio shilonii* *Xanthomonas*
perforans *Vibrio splendidus* *Zobellia* *Xanthomonas phaseoli* *Vibrio tubiashii* *Zobellia*
galactanivorans *Xanthomonas pisi* *Vibrio vulnificus* *Zobellia uliginosa* *Xanthomonas populi*
Zoogloea *Xanthomonas theicola* *Zoogloea ramigera* *Xanthomonas* *Zoogloea resiniphila*
translucens *Xanthomonas vesicatoria* *Xylella* *Xylella fastidiosa* *Xylophilus* *Xylophilus ampelinus*
Xenophilus *Yangia* *Yersinia mollaretii* *Zooshikella* *Zobellella* *Xenophilus azovorans* *Yangia*
pacifica *Yersinia philomiragia* *Zooshikella ganghwensis* *Zobellella denitrificans* *Xenorhabdus*
Yaniella *Yersinia pestis* *Zunongwangia* *Zobellella taiwanensis* *Xenorhabdus beddingii* *Yaniella*
flava *Yersinia pseudotuberculosis* *Zunongwangia profunda* *Xenorhabdus bovienii* *Yaniella*
halotolerans *Yersinia rohdei* *Zymobacter* *Zeaxanthinibacter* *Xenorhabdus cabanillasii* *Yeosuana*
Yersinia ruckeri *Zymobacter palmae* *Zeaxanthinibacter* *Xenorhabdus doucetiae* *Yeosuana*
aromativorans *Yokenella* *Zymomonas enoshimensis* *Xenorhabdus griffiniae* *Yersinia* *Yokenella*
regensburgei *Zymomonas mobilis* *Zhihengliuella* *Xenorhabdus hominickii* *Yersinia aldovae*
Yonghaparkia *Zymophilus* *Zhihengliuella* *Xenorhabdus koppenhoeferi* *Yersinia bercovieri*
Yonghaparkia alkaliphila *Zymophilus paucivorans* *halotolerans* *Xenorhabdus nematophila* *Yersinia*
enterocolitica *Zavarzinia* *Zymophilus raffinosivorans* *Xylanibacterium* *Xenorhabdus poinarii*
Yersinia entomophaga *Zavarzinia compransoris* *Xylanibacterium ulmi* *Xylanibacter* *Yersinia*
frederiksenii *Xylanibacter oryzae* *Yersinia intermedia* *Yersinia kristensenii*

(390) TABLE-US-00002 TABLE 2 Sequences Nucleic acid sequences herein are written in 5' to 3' direction; amino acid sequences are written in N- to C-terminal direction. SEQ ID NO: 1 (P10)
TTTCAATTTAATCATCCGGCTCGTATAATGTGTGGA SEQ ID NO: 2 (BCD14)
GGGCCCCAAGTTCACTTAAAAAGGAGATCAACAATGAAAGCAATTTTCGTA
CTGAAACATCTTAATCATGCGGTGGAGGGTTTCTAATG SEQ ID NO: 3 (gfp)
ATGAGCAAAGGAGAAGAAGAACTTTTCACTGGAGTTGTC SEQ IDs NO: 4 & 29
(example Expression Operating Unit, EOU) The EOU is (in 5' to 3' direction):-
[SEQ ID NO: 4]-[promoter]-[TIS]-[GFP-encoding nucleotide sequence]-[SEQ ID NO: 29] Where SEQ ID NO: 4 is
GAATTCAAAAGATCTTAAGTAAGTAAGAGTATACGTATATCGGCTAATAA
CGTATGAAGGCGCTTCGGCGCCTTTTTTTTATGGGGGTATTTTCATCCCAA
TCCACACGTCCAACGCACAGCAAACACCACGTCGACCCTATCAGCTGCGT
GCTTTCTATGAGTCGTTGCTGCATAACTTGACAATTAATCATCCGGCTCG
TATAATGTGTGGAA SEQ ID NO: 29 is
GGATCCAAACTCGAGTAAGGATCTCCAGGCATCAAATAAAACGAAAGGCT
CAGTCGAAAGACTGGGCCTTTCGTTTTATCTGTTGTTTGTGCGGTGAACGC
TCTCTACTAGAGTCACACTGGCTCACCTTCGGGTGGGCCTTTCTGCGTTT ATA SEQ
ID NO: 5 (Example Shine Dalgarno Sequence) AAAGAGGAGAAA SEQ ID
NO: 26 (Spacer sequence) CTTTGCCGCGCGCTTCGTCACGTAATTCTCGTCGCAA
SEQ ID NO: 27 (Spacer sequence)
GTTTGGCGATGGCGCGGGTGTGTTGTGCTTCGGCGT SEQ ID NO: 28 (Spacer
sequence) TGGGATGCCTACCGCAAGCAGCTTGGCCTGAA

(391) TABLE-US-00003 TABLE 3 Anderson Promoter Collection SEQ ID Measured NO:
Identifier Sequence.sup.a Strength.sup.b 6 BBa J23119
TTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGC n/a 7 BBa J23100
TTGACGGCTAGCTCAGTCCTAGGTACAGTGCTAGC 1 8 BBa J23101
TTTACAGCTAGCTCAGTCCTAGGTATTATGCTAGC 0.7 9 BBa J23102
TTGACAGCTAGCTCAGTCCTAGGTACTGTGCTAGC 0.86 10 BBa J23103
CTGATAGCTAGCTCAGTCCTAGGGATTATGCTAGC 0.01 11 BBa J23104

TTGACAGCTAGCTCAGTCCTAGGTATTGTGCTAGC 0.72 12 BBa J23105
 TTTACGGCTAGCTCAGTCCTAGGTACTATGCTAGC 0.24 13 BBa J23106
 TTTACGGCTAGCTCAGTCCTAGGTATAGTGCTAGC 0.47 14 BBa J23107
 TTTACGGCTAGCTCAGCCCTAGGTATTATGCTAGC 0.36 15 BBa J23108
 CTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGC 0.51 16 BBa J23109
 TTTACAGCTAGCTCAGTCCTAGGGACTGTGCTAGC 0.04 17 BBa J23110
 TTTACGGCTAGCTCAGTCCTAGGTACAATGCTAGC 0.33 18 BBa J23111
 TTGACGGCTAGCTCAGTCCTAGGTATAGTGCTAGC 0.58 19 BBa J23112
 CTGATAGCTAGCTCAGTCCTAGGGATTATGCTAGC 0 20 BBa J23113
 CTGATGGCTAGCTCAGTCCTAGGGATTATGCTAGC 0.01 21 BBa J23114
 TTTATGGCTAGCTCAGTCCTAGGTACAATGCTAGC 0.1 22 BBa J23115
 TTTATAGCTAGCTCAGCCCTTGGTACAATGCTAGC 0.15 23 BBa J23116
 TTGACAGCTAGCTCAGTCCTAGGGACTATGCTAGC 0.16 24 BBa J23117
 TTGACAGCTAGCTCAGTCCTAGGGATTGTGCTAGC 0.06 25 BBa J23118
 TTGACGGCTAGCTCAGTCCTAGGTATTGTGCTAGC 0.56 .sup.aalso shown in the Anderson Catalog, see parts.igem.org/Promoters/Catalog/Anderson bStrength is the Anderson Score (AS), e.g., a strength of 1 is a AS of 1. Reported activities of the promoters are given as the relative fluorescence of plasmids in strain TG1 grown in LB media to saturation. A suitable plasmid is EX-Ptet-S-rbsRFP-P "RFP reporter" as described at parts.igem.org/Part:BBa_J61002; insertion of a promoter element between XbaI and SpeI sites results in a RFP reporter.

Claims

1. A production strain bacterial cell comprising a nucleic acid vector for introduction into a target bacterial host cell for expression of Type I Cas3 and Cascade proteins in the target bacterial host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more cognate Cascade proteins, wherein the first nucleotide sequence is under the control of a promoter for controlling the expression of Type I Cas3 in the target bacterial host cell, wherein the promoter has a strength that is weaker than the Anderson Score strength of promoter BBa_J23108, wherein the nucleic acid vector further comprises: (i) a CRISPR array for producing a crRNA in the target bacterial host cell; or (ii) a nucleotide sequence encoding a guide RNA (gRNA), wherein the crRNA or gRNA comprises a spacer sequence complementary to a target sequence of the target bacterial host cell, and wherein the production strain bacterial cell does not comprise a crRNA or gRNA operable with the Cas3 to target and cut a chromosomal sequence of the production strain cell.
2. The production strain bacterial cell of claim 1, wherein the target bacterial host cell is selected from the group consisting of *Campylobacter*, *Burkholderia*, and *Acinetobacter*.
3. The production strain bacterial cell of claim 2, wherein the promoter is operable in a target host cell selected from the group consisting of *Campylobacter jejuni*, *Campylobacter coli*, *Campylobacter lari*, *Burkholderia cepacia*, *Burkholderia cenocepacia*, *Burkholderia multivorans*, *Burkholderia vietnamiensis*, *Burkholderia stabilis*, *Burkholderia thailandensis*, *Acinetobacter baumannii*, *Acinetobacter calcoaceticus*, *Acinetobacter haemolyticus*, *Acinetobacter lwoffii*, and *Acinetobacter schindleri*.
4. The production strain bacterial cell of claim 3, wherein the Type I Cas3 and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus*, or *B. halodurans* Cas and Cascade proteins.
5. The production strain bacterial cell of claim 1, wherein the nucleic acid vector comprises an operon for expression of the Type I Cas3 and Cascade proteins, and: (a) the first nucleotide sequence is between the promoter and the second nucleotide sequence in the operon; (b) the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence; or (c) the operon comprises, in 5' to 3' direction, the promoter, the first nucleotide

sequence, and the second nucleotide sequence.

6. The production strain bacterial cell of claim 1, wherein the promoter is a constitutive promoter.

7. The production strain bacterial cell of claim 1, wherein the promoter is repressible.

8. The production strain bacterial cell of claim 1, wherein the promoter has a strength that is greater than the Anderson Score strength of promoter BBa_J23114.

9. The production strain bacterial cell of claim 1, wherein the nucleic acid vector further comprises an origin of replication that is operable in the target bacterial host cell.

10. The production strain bacterial cell of claim 1, wherein the nucleic acid vector is devoid of a Cas adaption module.

11. The production strain bacterial cell of claim 1, wherein the nucleic acid vector is devoid of a nucleotide sequence encoding one or more of a Cas1, Cas2, Cas4, Cas6, Cas7, and Cas8.

12. The production strain bacterial cell of claim 1, wherein the second nucleotide sequence encodes one or more of (a)-(g): (a) Cas11, Cas7, and Cas8a1; (b) Cas8b1, Cas7, and Cas5; (c) Cas5, Cas8c, and Cas7; (d) Cas8U2, Cas7, Cas5, and Cas6; (e) Cas10d, Cas7, and Cas5; (f) Cas8e, Cas11, Cas7, Cas5, and Cas6; and (g) Cas8f, Cas5, Cas7, and Cas6f.

13. The production strain bacterial cell of claim 12, wherein the Type I Cas3 is a Cas3' or Cas3''.

14. The production strain bacterial cell of claim 12, wherein the Type I Cas3 is a Cas3, Cas3' or Cas3'', and wherein the Type I Cas3 is between the promoter and the second nucleotide sequence.

15. The production strain bacterial cell of claim 14, wherein the nucleic acid vector is devoid of a nucleotide sequence encoding a further Cas between the promoter and the Type I Cas3.

16. The production strain bacterial cell of claim 12, wherein the vector comprises the CRISPR array, the CRISPR array is cognate with the Type I Cas3, and wherein: (a) the CRISPR array is a Type IA array and the nucleic acid vector comprises Cas11, Cas7, and Cas8a1; (b) the CRISPR array is a Type IB array and the nucleic acid vector comprises Cas8b1, Cas7, and Cas5; (c) the CRISPR array is a Type IC array and the nucleic acid vector comprises Cas5, Cas8c, and Cas7; (d) the CRISPR array is a Type IU array and the nucleic acid vector comprises Cas8U2, Cas7, Cas5, and Cas6; (e) the CRISPR array is a Type ID array and the nucleic acid vector comprises Cas10d, Cas7, and Cas5; (f) the CRISPR array is a Type IE array and the nucleic acid vector comprises Cas8e, Cas11, Cas7, Cas5, and Cas6; or (g) the CRISPR array is a Type IF array and the nucleic acid vector comprises Cas8f, Cas5, Cas7, and Cas6f.

17. The production strain bacterial cell of claim 1, wherein the Type I Cas3 and Cascade are: (a) Type IA Cas and Cascade proteins; (b) Type IB Cas and Cascade proteins; (c) Type IC Cas and Cascade proteins; (d) Type ID Cas and Cascade proteins; (e) Type IE Cas and Cascade proteins; (f) Type IF Cas and Cascade proteins; or (g) Type IU Cas and Cascade proteins.

18. The production strain bacterial cell of claim 1, wherein the Type I Cas3 and Cascade are *E. coli* Cas and Cascade proteins.

19. The production strain bacterial cell of claim 1, wherein the Type I Cas3 and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus*, or *B. halodurans* Cas and Cascade proteins.

20. The production strain bacterial cell of claim 1, wherein the Type I Cas3 is a Cas3 of a CRISPR/Cas locus of *E. coli*, and wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter in *E. coli* is further than the distance between the Cas3-encoding sequence and the promoter for controlling the expression of Type I Cas3 in the nucleic acid vector.

21. The production strain bacterial cell of claim 1, wherein the CRISPR array or the gRNA-encoding sequence is under the control of a second promoter that is different from the promoter that controls the expression of the Type I Cas3.

22. The production strain bacterial cell of claim 1, wherein the nucleic acid vector is a plasmid or phagemid.

23. The production strain bacterial cell of claim 1, wherein the production strain bacterial cell comprises a nucleotide sequence whose expression is inducible to produce phage coat proteins in the cell of the production strain, wherein the production strain bacterial cell comprises amplified

copies of the nucleic acid vector, wherein the production strain bacterial cell is capable of packaging the amplified copies of the nucleic acid vector into phage particles or non-self-replicative transduction particles for introducing the amplified copies of the nucleic acid vector into the target host cell.

24. The production strain bacterial cell of claim 23, wherein the nucleic acid vector is a plasmid or phagemid and the delivery vehicle is a non-replicative transduction particle.

25. The production strain bacterial cell of claim 1, wherein the second nucleotide sequence is under the control of the same promoter as the first nucleotide sequence.

26. The production strain bacterial cell of claim 1, wherein the target sequence of the target bacterial host cell is a chromosomal sequence of the target bacterial host cell.

27. The production strain bacterial cell of claim 1, wherein the production strain bacterial cell is an *Escherichia coli* (*E. coli*) cell.
